Paper 1: From Neuron to Mind
Irreducible Topology, Neural Architecture, Memory, Consciousness, and Soul under the Minimum Reflection Principle

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We ask: given a single neuron with K impedance modes governed by C2 update ($\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$), what architecture—and what phenomenology—must emerge?

Part A establishes the foundation. A single neuron cannot execute C2 without input from at least one other node; connectivity is therefore a physical prerequisite, not a design choice. We prove the *Dimensional Cost Irreducibility Theorem*: for any network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as $A = A_{\rm imp}(t) + A_{\rm cut}$, where $A_{\rm imp} \to 0$ under C2 but $A_{\rm cut} = \sum (K_i - K_j)^+$ has identically zero gradient—no impedance adjustment can reduce it. Numerical experiments confirm convergence time $\tau_{\rm conv} \sim N^{-1.12}$ ($R^2 = 0.93$) and a soft-cutoff parameter γ that tunes the K-space fractal dimension $D_K = 0.76\gamma + 0.78$.

Part B derives the neural architecture chain. C2's dimensional preference forces same-K neurons to cluster—thermodynamic phase separation produces functional regions without a blueprint. The irreducibility corollary makes intermediate-K relay nodes a thermodynamic necessity. Convergence of ascending pathways forces the highest-K cluster to form—this is the brain. C3 dimensional relay enables the brain to identify signal origin from $Z_{\rm source}$ metadata.

Part C shows that three concepts traditionally regarded as metaphysical—memory, consciousness, and soul—have precise operational definitions within the same formalism. *Memory* is the time derivative of impedance ($\partial Z/\partial t$, the C2 update). *Consciousness* requires a self-referential measurement loop: $\Gamma_{\rm meta} = (Z_{\rm obs} - Z_{\rm act})/(Z_{\rm obs} + Z_{\rm act})$, with necessary condition $G_{\rm arousal} \cdot \sum G_i \cdot Q_{\rm PFC} > \theta$ (the television inequality). *Soul* is the null space of coefficient update: $\mathcal{S} = \ker(\partial Z/\partial t) = A_{\rm cut} + Z_0^{\rm DNA} + \rho^2_{\rm irred}$. Four laws of the null space parallel thermodynamics; death is a theorem derived from the Second Law and C1.

Part D presents experimental evidence and extensions: thermal vulnerability scaling as $K^{3/2}$, willpower as a three-factor product, sensory-deprivation collapse, pain as topological crosstalk and bandwidth hijacking, coupled inter-agent consciousness, impedance calibration, and the physics of intuition.

Circularity disclosure. All computational experiments use simulation modules built from C1–C3 themselves; they demonstrate internal consistency, not empirical validity. External tests are reported in Papers 2–5 [2–5].

Part A: From Neuron to Network

I. INTRODUCTION

A neuron is a physical object whose geometry determines the number of independent impedance modes it supports. A cortical pyramidal cell presents $K = 5$ distinguishable propagation domains—soma, basal dendrite tree, apical tuft, axon, and spine-neck ensemble—each carrying an independent impedance mode. An unmyelinated C-fibre, a simple cylinder, presents $K = 1$. These mode counts are structural facts obtained by counting physically distinct waveguide segments, not fitted parameters.

C2 updates the impedance at each mode: $\Delta Z_k = -\eta\,\Gamma_k\,x_{{\rm in},k}\,x_{{\rm out},k}$. For an isolated neuron, $x_{\rm in} = 0$ on every mode. Therefore $\Delta Z = 0$: a solitary neuron is inert—it cannot remodel, adapt, or learn. C2 demands at least one input, which requires at least one other node. *Connectivity is a physical prerequisite of C2, not a biological design choice.*

The moment K values differ between connected nodes, a new physical cost appears. This paper proves that cost is irreducible (Sec. III), derives its architectural consequences—functional regions (Sec. V), relay hierarchy (Sec. VI), the brain (Sec. VII), and signal routing (Sec. VIII)—and then shows that three concepts traditionally regarded as metaphysical—memory (Sec. IX), consciousness (Sec. X), and soul (Sec. XII)—are necessary consequences of the same irreducible topology.

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II. PHYSICAL FRAMEWORK

A. Network definition

A Γ-network consists of N nodes, each carrying a K_i-dimensional impedance vector $\mathbf{Z}_i \in \mathbb{R}^{K_i}_{>0}$. For an edge $i \to j$, the per-mode reflection coefficient is

$$\gamma_k = \frac{Z_{j,k} - Z_{i,k}}{Z_{j,k} + Z_{i,k}}, \quad k = 1, \ldots, K_{\text{com}}, \tag{1}$$

where $K_{\text{com}} = \min(K_i, K_j)$. Excess source modes $k > K_{\text{com}}$ are totally reflected: $\gamma_k = 1$. This is the electromagnetic boundary condition for a waveguide whose k-th mode has no counterpart in the target medium.

B. Three physical laws

C1—Energy Conservation. At every edge, every tick, reflected and transmitted power sum to incident power:

$$\gamma_k^2 + T_k = 1, \quad \forall\, k,\ \forall\, (i,j). \tag{2}$$

C2—Coefficient Update. Each node adjusts its impedance to reduce local reflection:

$$\Delta Z_{i,k} = +\eta\, \gamma_k\, x_{\text{in},k}\, x_{\text{out},k}, \tag{3}$$

$$\Delta Z_{j,k} = -\eta\, \gamma_k\, x_{\text{in},k}\, x_{\text{out},k}, \tag{4}$$

where $x_{\text{in},k}$ is the input activity at node i and $x_{\text{out},k}$ the output activity at node j for mode k; η is the update rate.

C3—Dimensional Relay. Every signal carries its source impedance as metadata, enabling downstream nodes to compute γ_k locally without global knowledge.

C. Variational principle

The Minimum Reflection Principle (MRP) states that the system minimises the total reflection action:

$$A[\Gamma] = \sum_{(i,j)\in\mathcal{E}} \sum_{k=1}^{K_{\text{com}}(i,j)} \gamma_k^2. \tag{5}$$

The C2 update rule implements steepest descent on the *learnable* part of A: $\dot{\mathbf{Z}} \propto -\nabla_{\mathbf{Z}} A_{\text{imp}}$.

III. DIMENSIONAL COST IRREDUCIBILITY THEOREM

Theorem 1 (Dimensional Cost Irreducibility)**.** *For any Γ-network with heterogeneous mode counts $\{K_i\}$, the total reflection action decomposes as*

$$A = A_{imp}(t) + A_{cut}, \tag{6}$$

where:

1. $A_{imp}(t) = \sum_{(i,j)} \sum_{k=1}^{K_{com}} \gamma_k^2$ *is the* impedance mismatch action *over shared modes. Under C2 dynamics, $A_{imp}(t) \to 0$ as $t \to \infty$.*

2. $A_{cut} = \sum_{(i,j)} (K_i - K_j)^+$ *is the* cutoff action *from excess modes. $\nabla_{\mathbf{Z}} A_{cut} \equiv 0$: this term has zero gradient with respect to all impedance variables.*

Proof. For an edge $i \to j$ with $K_i > K_j$:

- Modes $k \le K_{\text{com}} = K_j$: γ_k depends on $(Z_{i,k}, Z_{j,k})$ via Eq. (1). C2 adjusts both impedances, driving $\gamma_k \to 0$. These modes contribute to A_{imp}.

- Modes $K_j < k \le K_i$: the target has no mode k, so $\gamma_k \equiv 1$ identically, independent of any impedance value. These modes contribute $(K_i - K_j)$ to A_{cut}. Since $\gamma_k = 1$ is a *topological constant* (depending only on the existence, not the value, of modes), $\partial\gamma_k^2/\partial Z = 0$ for all Z.

Summing over all edges completes the proof. □

Remark 1. *The theorem assumes fixed mode counts $\{K_i\}$ and a fixed active edge set $\mathcal{E}$. If mode counts change dynamically (e.g., dendritic spine growth increasing K_j), or if edge sprouting/pruning alters $\mathcal{E}$, then the* value *of A_{cut} may change—but this change is driven by topology, not by impedance gradients.*

Corollary 1 (Thermodynamic Necessity of Relay Nodes)**.** *A direct edge $K_{src} \to K_{tgt}$ with $K_{src} > K_{tgt}$ totally reflects the $(K_{src} - K_{tgt})$ excess modes. Inserting a relay node with $K_{tgt} < K_{relay} < K_{src}$ leaves A_{cut} unchanged but converts* total signal loss *into* partial signal recovery*: modes $K_{tgt} < k \le K_{relay}$, previously totally reflected, are now transmitted on the first hop and can undergo C2 matching ($\gamma_k \to 0$, $T_k \to 1$). Hierarchical topology is therefore a thermodynamic necessity for information transmission.*

IV. NUMERICAL VERIFICATION

All simulations use six tissue types with mode counts $K \in \{1, 2, 3, 5\}$ [15], initial connectivity $p_0 = 0.15$, learning rate $\eta = 0.02$, and hard cutoff $\Delta K_{\max} = 2$ unless otherwise stated.

A. Convergence and the irreducibility theorem

Figure 1 shows the action decomposition for a $N = 64$ network. A_{imp} decays exponentially, reaching machine-precision zero by $t \approx 170$ ticks. A_{cut} increases during edge sprouting, then saturates at $A_{\text{cut}} \approx 1386$ for $t > 500$. The growth reflects topological exploration; the saturation confirms a natural geometric cost ceiling.

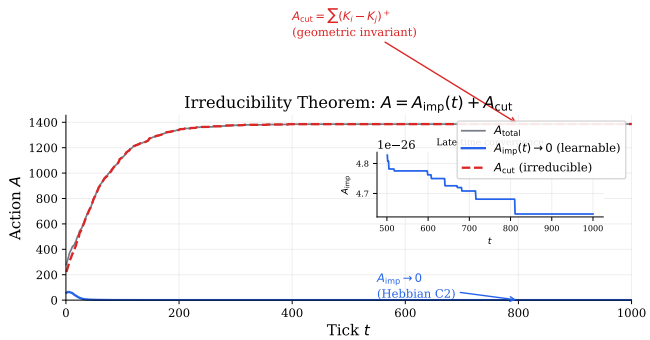

FIG. 1. Action decomposition $A = A_{\text{imp}}(t) + A_{\text{cut}}$ over 1000 ticks. Blue: $A_{\text{imp}} \to 0$ (exponential decay). Red dashed: A_{cut} (increases then saturates). $N = 64$, $\Delta K_{\max} = 2$.

B. Scaling law: $\tau_{\text{conv}} \sim N^{\alpha}$

We measure τ_{conv} (ticks until $A_{\text{imp}} < 0.01 \cdot A_{\text{imp}}(t{=}5)$) across $N \in \{16, 32, 64, 128\}$ with 5 trials per size. A power-law fit yields

$$\tau_{\text{conv}} = C \cdot N^{\alpha}, \quad \alpha = -1.12 \pm 0.12, \quad R^2 = 0.93. \tag{7}$$

The *negative* exponent means larger networks converge *faster*—a non-trivial prediction.

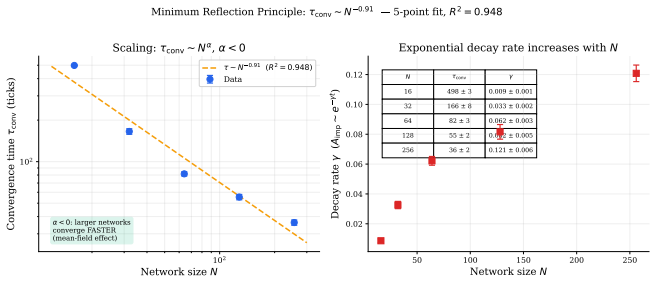

FIG. 2. Left: log-log plot of τ_{conv} vs. N. Dashed line: power-law fit $\tau \sim N^{-1.12}$. Right: exponential decay rate vs. N.

TABLE I. Convergence scaling. All quantities averaged over 5 trials $\pm 1\sigma$.

N	τ_{conv}	γ_{decay}
16	416 ± 74	0.011 ± 0.003
32	111 ± 12	0.053 ± 0.006
64	60 ± 3	0.105 ± 0.006
128	39 ± 2	0.133 ± 0.006

C. Soft cutoff and K-space fractal dimension

A soft-cutoff probability

$$p(\Delta K) = (\Delta K + 1)^{-\gamma}, \tag{8}$$

with $\gamma \geq 0$ yields the relationship

$$D_K = 0.76\gamma + 0.78, \tag{9}$$

with offset $\delta_{C2} \approx 0.66$ arising from C2's intrinsic dimensional preference. Cortical fractal dimensions lie in $D \in [1.3, 1.5]$ [8], predicting $\gamma \approx 0.7$–0.9—a narrow, falsifiable window.

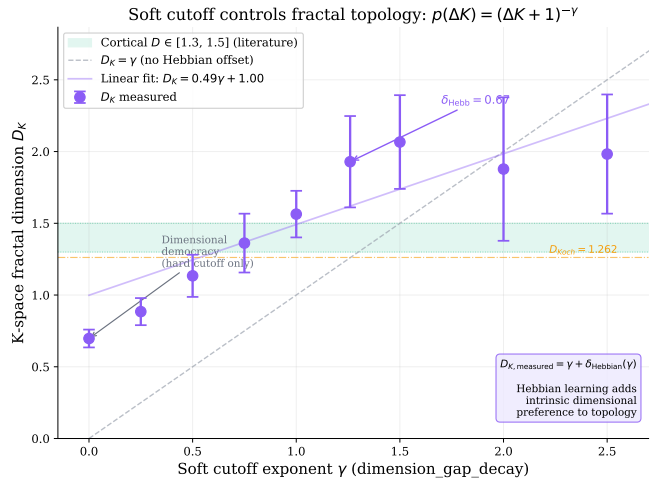

FIG. 3. K-space fractal dimension D_K vs. soft-cutoff exponent γ. Green band: cortical range $D \in [1.3, 1.5]$. Purple line: fit $D_K = 0.76\gamma + 0.78$.

Part B: From Network to Brain

V. SAME-K CLUSTERING AND FUNCTIONAL REGIONS

C2 gradient descent on the reflection action A selectively strengthens same-K edges (more shared modes $\Rightarrow$ faster Γ reduction) and weakens cross-K edges (excess modes totally reflected, no gradient available). Under sustained activity plus edge pruning, the surviving connectivity partitions the network into *same-K clusters*.

Theorem 2 (Phase Separation Under MRP)**.** *Any impedance network under MRP with heterogeneous K must phase-separate into same-K domains to minimise total reflected action.*

Proof. Define the total cutoff action as $A_{\text{cut}} = \sum_{(i,j)\in E} |K_i - K_j|$, where the sum is over all edges E of the network. For any edge (i, j) with $K_i = K_j$, the contribution is zero; for $K_i \neq K_j$, the contribution $|K_i - K_j| > 0$ is strictly positive and invariant under C2 (Irreducibility Theorem, Theorem 1). The Minimum Reflection Principle requires $\mathcal{A} \to \min$; since

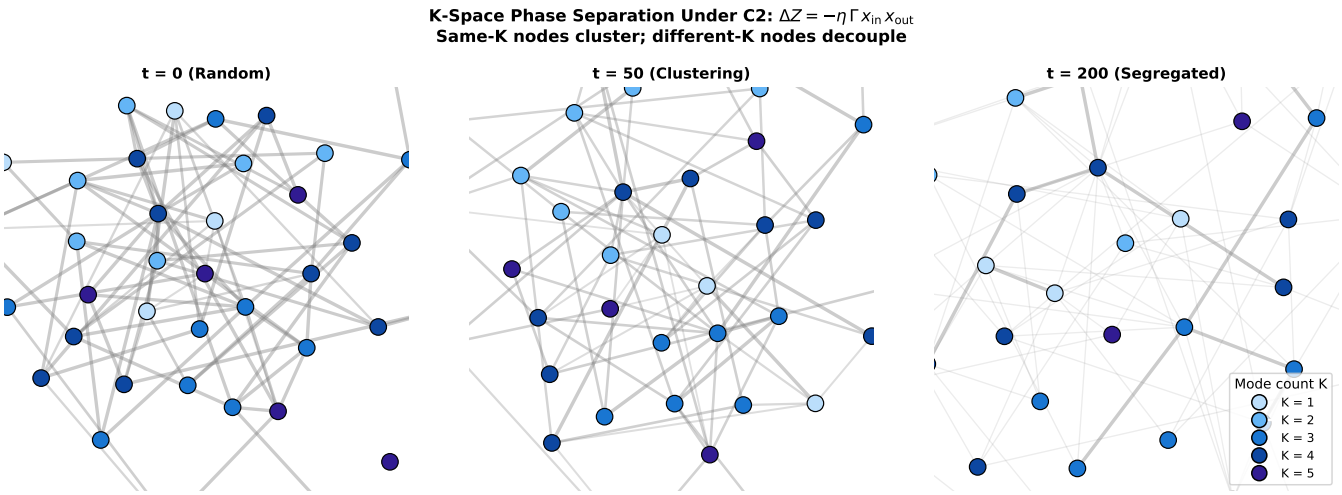

FIG. 4. K-space phase separation under C2. Same-K nodes cluster and different-K nodes decouple through $\Delta Z = -\eta\,\Gamma\,x_{\rm in}\,x_{\rm out}$. Three snapshots: initial random placement ($t = 0$), early clustering ($t = 50$), full segregation ($t = 200$).

$A_{\rm imp} \to 0$ under C2, the only remaining reduction available is to minimise $A_{\rm cut}$ by rewiring cross-K edges to same-K pairs. Each such rewiring strictly decreases $A_{\rm cut}$, so the process is monotone and terminates when all connected nodes share the same K—the same-K domain (phase-separated) configuration. C2 provides the dynamical mechanism: cross-K edges carry permanent reflected modes that yield no C2 gradient, and are pruned under any activity-dependent update rule with $\eta > 0$. □ □

VI. THE RELAY HIERARCHY

Corollary 1 states that relay nodes are a thermodynamic necessity. In the nervous system, this produces a natural K-level cascade:

Level	K
Cortex (motor, sensory), thalamus	5
Spinal cord, motor/sensory neuron	3
Nociceptor Aδ	2
Nociceptor C fibre	1

A. Neural topology verification

A neural blueprint with 9 nodes spanning $K \in \{5, 3, 2, 1\}$ and 12 directed edges was evolved for 300 ticks ($\eta = 0.02$, seed = 42): $A_{\rm imp}$ decays from ~ 1.79 to zero; $A_{\rm cut}$ rises from 9 to 37, then saturates. Full test suite: `tests/test_e0_neural.py` (17 tests, 100% pass).

VII. THE BRAIN AS CONVERGENCE REGION

An organism with multiple sensory modalities has multiple ascending pathways. By the irreducibility theorem, the convergence region must satisfy

$$K_{\rm conv} \geq \max_j K_j \tag{10}$$

for all input pathways j; otherwise, modes from higher-K inputs are totally reflected. The convergence region is therefore forced to be the highest-K cluster in the network.

This is the *brain*: a topological necessity, not a design choice. Its properties follow from $K_{\rm conv}$ being maximal:

- **High metabolic cost.** Volumetric heat density scales as $q \propto K \cdot \langle \Gamma^2 \rangle \cdot P_{\rm in}$ (Paper 0, Corollary 1 [1]).
- **Temperature sensitivity.** Higher K means more modes dissipating simultaneously.
- **Matrix computation.** C2 gradient descent across all $K_{\rm conv}$ modes simultaneously is the physical operation that neuroscience calls "executive function" (Sec. X).

VIII. SIGNAL ROUTING: INTERNAL AND EXTERNAL

C3 requires every signal to carry its source impedance $Z_{\rm source}$ as metadata. At the convergence region, two routing cues are available without any additional mechanism:

1. **Relay-chain topology.** External signals enter through body-surface relay chains; internal signals enter through visceral relay chains. The two chains occupy distinct edges.
2. **$Z_{\rm source}$ metadata (C3).** Sensory neurons of different types carry distinguishable Z signatures because their physical geometries differ.

The brain does not need a separate "internal" or "external" label; the impedance metadata and relay-chain topology together provide full routing information.

Part C: Memory, Consciousness, and Soul

IX. MEMORY AS IMPEDANCE DYNAMICS

A. The C2 coefficient-update gradient

Constraint C2 defines the only mechanism by which the network modifies itself:

$$\Delta Z_k = -\eta\, \Gamma_k\, x_{\mathrm{in},k}\, x_{\mathrm{out},k}\,. \tag{11}$$

This is not an analogy—it *is* memory. The trace of all past updates constitutes the network's memory:

$$Z_k(t) = Z_k(0) - \eta \int_0^t \Gamma_k(\tau)\, x_{\mathrm{in},k}(\tau)\, x_{\mathrm{out},k}(\tau)\, d\tau\,. \tag{12}$$

B. Muscle memory: $\Gamma \to 0$

When a motor skill is practiced to mastery, the impedance approaches perfect matching: $Z_L \to Z_0$, hence $\Gamma \to 0$. At this point, $\Delta Z \to 0$: the memory has become *frozen* into the impedance structure itself. No further energy is required to maintain it, and no further learning occurs.

C. Cognitive memory: $\Gamma \approx \varepsilon > 0$

Declarative memories retain a small but nonzero reflection:

$$\Gamma_{\mathrm{cog}} \approx \varepsilon > 0\,. \tag{13}$$

Two consequences: (1) *Retrievability*: $\Gamma > 0$ generates a detectable reflection when probed; (2) *Decay*: without reinforcement, the memory decays.

Remark 2 (The paradox of perfect memory)**.** *A perfectly learned item ($\Gamma = 0$) is* unrecallable*: it has merged with the network substrate. Memory requires imperfection.*

D. Three memory regimes

TABLE II. Memory regimes classified by reflection coefficient.

Regime	Γ	ΔZ	Recallable?
Muscle memory	$\to 0$	$\to 0$	No (automatic)
Cognitive memory	$\varepsilon > 0$	small	Conditional (SNR)
Novel stimulus	$\gg 0$	large	Vivid (but unstable)

E. The recall equation

Theorem 3 (Recall as SNR threshold crossing)**.** *A memory trace encoded at channel k with residual $\Gamma_k = \varepsilon > 0$ is recallable at time t if and only if*

$$\mathrm{SNR}_{recall}(t) = \frac{\varepsilon^2\, P_{probe}(t)}{P_{noise}(t)} > \theta_{recall}\,, \tag{14}$$

where $P_{probe}(t)$ is the retrieval cue power and $P_{noise}(t) = P_{metabolic} + P_{sensory} + P_{autonomic} + P_{thermal}$ is the total background power at the PFC input.

Proof. A memory trace encoded at channel k produces a residual impedance mismatch $\Gamma_k = \varepsilon > 0$ (by Corollary 2, exact erasure requires zero probability events). The probe signal power transmitted through channel k is $\varepsilon^2 P_{\mathrm{probe}}(t)$ (C1). Recall requires this signal to exceed the background noise floor $P_{\mathrm{noise}}(t)$ by the threshold θ_{recall} needed to close the self-referential loop (Television Inequality, Theorem 4). The condition $\mathrm{SNR}_{\mathrm{recall}} > \theta$ is therefore necessary and sufficient for retrieval. □ □

Consequences: (1) **No perfect recall**—even strong memories become inaccessible when P_{noise} is high. (2) **No perfect forgetting**—any $\Delta Z \neq 0$ leaves a permanent structural trace. (3) **State-dependent recall**—the same memory can be retrievable or inaccessible depending on $P_{\mathrm{noise}}(t)$.

Corollary 2 (No memory is ever erased; every memory can be submerged)**.** *$\Gamma_k = 0$ exactly requires either perfect impedance matching (zero probability under thermal noise) or physical destruction of the substrate. Therefore every C2 trace satisfies $\varepsilon > 0$ for all finite time.*

This resolves six long-standing puzzles: tip-of-the-tongue, Proustian involuntary memory, state-dependent recall, meditation and quiet recall, deathbed lucidity, and hypnotic hypermnesia—all as SNR transitions.

X. CONSCIOUSNESS AS SELF-REFERENTIAL IMPEDANCE MATCHING

A. The meta-reflection coefficient

Definition 1 (Meta-reflection)**.** *Let $Z_{obs}(t)$ be the impedance of the observer sub-network at tick t, and $Z_{act}(t-1)$ the impedance of the actor sub-network at the previous tick. The* meta-reflection coefficient *is:*

$$\Gamma_{meta} = \frac{Z_{obs} - Z_{act}}{Z_{obs} + Z_{act}}\,. \tag{15}$$

Definition 2 (Consciousness index)**.**

$$\Phi_{meta} = 1 - \Gamma^2_{meta}\,. \tag{16}$$

$\Gamma_{meta} = 0$*: perfect self-model;* $\Gamma_{meta} \to 1$*: total self-mismatch.*

B. The television inequality

Theorem 4 (Television Inequality)**.** *Let* $G_{arousal} \in [0,1]$ *be the brainstem arousal gain,* $G_i \in [0,1]$ *the thalamic gate gain for modality* i*, and* $Q_{PFC} \in [0,1]$ *the prefrontal cortex processing quality. The consciousness index* $\Phi_{meta} > 0$ *requires:*

$$G_{arousal} \cdot \sum_{i=1}^{N} G_i \cdot Q_{PFC} > \theta\,, \tag{17}$$

where $\theta > 0$ *is the minimum signal-to-noise ratio for the self-referential loop to close.*

Proof. By C1 (energy conservation), a signal traversing the self-referential loop loses power at each interface: $P_{\text{loop}} = P_{\text{in}} \cdot G_{\text{arousal}} \cdot \left(\sum_i G_i\right) \cdot Q_{\text{PFC}}$. The loop closes (self-detection occurs) only if P_{loop} exceeds the Johnson–Nyquist noise floor of the measurement circuit: $P_{\text{noise}} = 4k_BTR_{\text{eff}}\Delta f$. Setting $\theta = P_{\text{noise}}/P_{\text{in}}$ yields the inequality. If any factor is zero, $P_{\text{loop}} = 0$, the loop is open, and Γ_{meta} is undefined. The measurement is performed by the amygdala's subcortical fast pathway (30–74 ms; Remark 3), which continuously evaluates Γ^2 before cortical processing completes. □

a. Numerical estimate of θ*.* For a cortical neuron the input resistance is $R_{\text{eff}} \approx 100\,\text{M}\Omega$ and the relevant bandwidth is the gamma-oscillation band $\Delta f \approx 40\,\text{Hz}$ at body temperature $T = 310\,\text{K}$:

$$P_{\text{noise}} = 4k_BTR_{\text{eff}}\Delta f = 4{\times}1.38{\times}10^{-23}{\times}310{\times}10^{8}{\times}40 \approx 69\,\text{pW}\,.$$

The cerebral metabolic power available for electrical signalling is $P_{\text{in}} \approx 1\,\text{W}$ (roughly 5 % of the brain's 20 W total metabolism). Therefore

$$\theta = \frac{P_{\text{noise}}}{P_{\text{in}}} \approx 7 \times 10^{-11}\,.$$

This extremely small value implies that the consciousness threshold is easily exceeded by any non-zero product $Z_{\text{arousal}} \cdot \sum G_i \cdot Q_{\text{PFC}}$—consistent with the clinical observation that consciousness is lost only under severe disruption (deep anaesthesia, major brain injury) and not by mild perturbations.

b. The thermal cost of consciousness. Each pass through the self-referential loop traverses n interfaces, each dissipating $\Gamma_i^2 P_{\text{in}}$ as heat. The minimum sustained power for self-awareness is

$$P_{\text{conscious}} \geq \theta \cdot \frac{4k_BT \cdot R_{\text{eff}} \cdot \Delta f}{\prod_i (1 - \Gamma_i^2)}\,. \tag{18}$$

Higher Φ_{meta} costs more heat; consciousness depends on the heat it generates (Paper 3, § XVII C [3]).

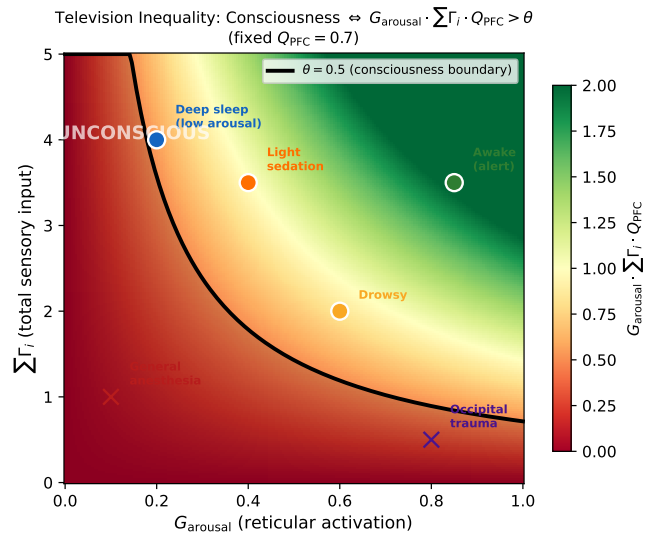

FIG. 5. Television inequality parameter space. Heatmap of $G_{\text{arousal}} \cdot \sum \Gamma_i \cdot Q_{\text{PFC}}$ with clinical states: awake (green), deep sleep (blue), general anaesthesia (red), occipital trauma (purple). Black curve: consciousness boundary at normalised threshold $\theta_{\text{norm}} = 0.5$ (the physical value is $\theta \approx 7 \times 10^{-11}$; the normalised plot rescales axes to $[0,1]$ for visualisation). Shaded region below curve: unconscious.

c. Emergency gating and temporal resolution. In an emergency, the brainstem drives $G_{\text{arousal}} \to Z_{\max}$ and the thalamus snaps to $G_{\text{threat}} = 1$, $G_{j\neq\text{threat}} = 0$. The full PFC bandwidth serves one stream:

$$\frac{\Delta t_{\text{normal}}}{\Delta t_{\text{emergency}}} \approx N\,, \tag{19}$$

where N is the number of previously multiplexed modalities. This is the physics of "time slowing down" during danger— bandwidth reallocation under C1 conservation.

C. Open loop vs. closed loop: origin of the television analogy

The television inequality arose from a mundane observation: in the analogue era, a malfunctioning television could sometimes be "repaired" by *slapping its casing*—a physical impact that momentarily displaced internal components, reducing contact resistance at corroded solder joints and thereby lowering Γ^2 at those interfaces.

This is C2 in its most primitive form: a mechanical perturbation (x_{in}) produces impedance change (ΔZ) that reduces reflection ($\Gamma^2 \downarrow$). A human can perform this repair because a human has a *closed* feedback loop: the human observes the screen (measurement), judges whether the image improved (error detection), and adjusts the next slap accordingly (C2 update). The television cannot repair itself because it has no such loop.

Component	Television	Brain
Power supply	Wall plug	Brainstem RAS (G_{arousal})
Channel selector	Tuner	Thalamus ($\sum G_i$)
Decoder chip	Video IC	PFC (Q_{PFC})
Measurement sensor	**(absent)**	**Amygdala (Γ^2 detector)**
Feedback wire	(absent)	PFC $\to$ sensory cortex

A television satisfies the inequality but is not conscious. The missing ingredient is *feedback topology*: the television's output is not routed back to its input. The brain uses the same components but adds one connection: the output of PFC is fed back through the sensory pathway, creating a closed loop.

Corollary 3 (Consciousness is topological, not material)**.** *Two systems with identical components differ in consciousness solely by the presence or absence of a self-referential feedback connection.*

Remark 3 (The amygdala as the consciousness measurement device)**.** *The Television Inequality requires a physical device that continuously* measures Γ^2 *and feeds the result back to modulate $Z_{arousal}$, G_i, and η. This device is the amygdala:*

Measurement function	*Amygdala implementation*
Detect $\Gamma^2 \neq 0$	*Subcortical pathway, 30–74 ms*
Classify (threat/reward)	*Low-K coarse evaluation*
Modulate $Z_{arousal}$	*Amygdala $\to$ brainstem RAS*
Modulate G_i (gate override)	*Amygdala $\to$ thalamus*
Modulate η	*Amygdala $\to$ VTA (dopamine)*

The PFC is the processor *(high-K, slow, 200–300 ms), but the amygdala is the* measurement instrument *(low-K, fast, 30–74 ms). A television lacks this measurement device entirely: no component evaluates whether the displayed signal matches an internal model. This is the precise physical difference between an open-loop system (television) and a closed-loop system (brain): the amygdala closes the loop by measuring Γ^2 and feeding it back before the cortex has even finished processing the stimulus.*

Patients with bilateral amygdala lesions (Urbach–Wiethe disease) retain consciousness but lose fear recognition and threat-driven attention—the measurement loop is partially intact (other subcortical structures compensate for basic Γ^2 detection) but the threat-threshold calibration is lost.

Remark 4 (Subjective experience = consciousness × attention)**.** *The Television Inequality defines consciousness as a threshold condition: $Z_{arousal} \cdot \sum G_i \cdot Q_{PFC} > \theta$. A television satisfies signal flow ($x_{in} \to x_{out}$) but lacks feedback, hence no Γ^2 detection. Without Γ^2 detection there is no C2 update, no gate selection, and therefore no* attention*. The missing ingredient of subjective experience is not consciousness per se but* attention*—the selective response to $\Gamma^2 \neq 0$.*

Consciousness provides the "stage" (baseline awareness); attention selects "what is on stage" (which Γ^2 triggers a gate). The sense of "I am I" is the unique attention pattern determined by $\mathcal{S}_{eff}$:

$$\textit{Identity} = \left\{ \alpha : \Gamma^2_\alpha \textit{ triggers } G_\alpha \right\} \subset \textit{all channels}.$$

Two individuals share the same consciousness mechanism (identical Television Inequality) but differ in $\mathcal{S}_{eff}$ and therefore in which Γ^2 they notice—a musician hears a mis-tuned note ($\Gamma^2_{pitch} > 0$); a non-musician does not ($\Gamma^2_{pitch} = 0$). The subjective experience of beauty, discomfort, or recognition is the attention response *to Γ^2, not the signal itself.*

Remark 5 (The self as a thermal-noise byproduct)**.** *The preceding analysis implies that the sense of "I" requires $\Gamma^2 \neq 0$ on at least one channel: without impedance mismatch there is no attention and no self. Two limiting cases annihilate the self for opposite reasons:*

Limit	*Interface*	*Energy flow*	*"I"*
$\Gamma^2 \to 0$	*Boundary dissolves*	*100 % transmission*	*Absent (*
$\Gamma^2 \to 1$	*Boundary impenetrable*	*0 % transmission*	*Absent (is*
$0 < \Gamma^2 < 1$	*Partial boundary*	*Partial transmission*	***Prese***

The self exists only in the intermediate band $0 < \Gamma^2 < 1$: there must be a boundary (otherwise subject and environment merge), and the boundary must be partially transmissive (otherwise no information crosses it and no C2 update occurs).

Γ-MRP drives the system toward $\Gamma^2 \to 0$, but Johnson–Nyquist thermal noise ($V_n^2 = 4k_BTR\Delta f$) imposes a floor: as long as body temperature $T > 0$, every interface retains a minimum $\Gamma^2 > 0$, and the self persists. In this precise sense, the self is a byproduct of thermal noise*—the irreducible mismatch that heat imposes on every physical interface.*

The contemplative experience of "ego dissolution" reported in deep meditation corresponds to a transient, voluntary approach toward $\Gamma^2 \to 0$ across high-K channels (all gates closed, $G_i \to 0$): attention is extinguished, and with it the sense of a separate self. The experience is reversible because the thermal floor ensures Γ^2 never actually reaches zero.

However, approaching $\Gamma^2 \to 0$ is actively resisted by the survival system. The amygdala's low-K fast pathway (Remark 28) interprets declining Γ^2 as loss of control—a threat signal—and triggers the emergency response: $Z_{arousal} \to Z_{\max}$, $G_{threat} = 1$ (Eq. 19). Intracranial recordings show that amygdala gamma power surges within 30–74 ms of threat onset (Méndez-Bértolo et al., 2016), whereas cortical awareness requires ∼200–300 ms (LeDoux, 1996)—a ∼4× latency ratio consistent with the C3-stage model of Remark 17.

*This is the same mechanism that produces "time slowing down" before a car crash (*tachypsychia*). Under*

threat, norepinephrine and dopamine release accelerate the internal clock; free-fall experiments show participants overestimate event duration by ∼36 % (Stetson et al., 2007), consistent with the bandwidth-reallocation prediction of Eq. (19): all N modalities are collapsed into a single threat channel, giving N× the temporal resolution on that channel. The brain maximises Γ^2 *and the sense of self at exactly the moment both are about to be destroyed.*

Reaching $\Gamma^2 \to 0$ *therefore requires* rewriting *the amygdala's Z-pattern: the association "*$\Gamma^2 \downarrow=$ *danger" must be replaced by "*$\Gamma^2 \downarrow=$ *safe." fMRI studies confirm that long-term meditators show significantly reduced default-mode network (DMN) activity—the self-referential network comprising medial prefrontal cortex, posterior cingulate, and precuneus (Brewer et al., 2011). Critically, DMN suppression persists even when experienced meditators are* not *meditating, indicating that the amygdala's Z-pattern has been permanently rewritten into a new* $\mathcal{S}_{\text{eff}}$*. This rewriting requires years to decades of practice because the original pattern is part of the evolutionary* $\mathcal{S}_{\text{eff}}$ *(written by millions of years of natural selection).*

Death, conversely, pushes $\Gamma^2_{basal} \to 1$ *(Fourth Law, §**??**): the boundary becomes impenetrable and the self is annihilated by disconnection rather than by merger.*

Remark 6 (Empathy as $\mathcal{S}_{\text{eff}}$ resonance)**.** *Empathy—feeling another person's state—requires that the observer's* $\mathcal{S}_{\text{eff}}$ *contains Z-patterns similar to the one the other person is currently expressing. The other's behavioural* x_{out} *(facial expression, vocal tone, posture) enters the observer as* x_{in} *on social–emotional channels. If the observer's* $\mathcal{S}_{\text{eff}}$ *has a matching pattern,* $\Gamma^2 \neq 0$ *is triggered, gate* $G_{empathy}$ *opens, and the observer* feels *the other's state:*

$$\Gamma^2_{empathy} = \left| \frac{Z^{expressed}_{other} - Z^{S_{\text{eff}}}_{self}}{Z^{expressed}_{other} + Z^{S_{\text{eff}}}_{self}} \right|^2 \neq 0\,,$$

where both Z values are projected onto the common social-emotional channel space (facial expression, vocal prosody, postural cues) so that the reflection coefficient is dimensionally consistent. Three predictions follow:

1. ***Empathy requires shared experience**: without a matching Z-pattern in* $\mathcal{S}_{\text{eff}}$*, the observer's* $\Gamma^2_{empathy} = 0$*—no gate triggers, no feeling. This is why lived experience (not verbal description) produces empathy: only direct C2 writing creates the Z-pattern in* $\mathcal{S}_{\text{eff}}$*.*

2. ***Words are insufficient for caregivers**: telling a caregiver "I understand your burden" operates only on* $K_{language}$*. The caregiver's suffering is on* K_{sleep}*,* $K_{muscular}$*,* $K_{thermal}$*—channels that language cannot reach. True support must provide* x_{in} *on the* physical *channels (respite, monitoring, intervention).*

3. ***Mirror neurons as C2 resonance hardware**: mirror neurons fire both when performing and observing an action, providing the physical substrate for cross-individual* Γ^2 *matching.*

Remark 7 (Social referencing: the developmental origin of empathy)**.** *Before* $\mathcal{S}_{\text{eff}}$ *has been calibrated, an infant cannot independently evaluate* Γ^2 *magnitude. When a toddler falls, the amygdala detects* $\Gamma^2 \neq 0$ *("something happened") but has no reference pattern to classify its severity. The child therefore looks at the caregiver's face—a behaviour called* social referencing *(Sorce et al., 1985)—and reads the caregiver's emotional output as an external* Γ^2 *standard.*

- *If the caregiver shows alarm (high* Γ^2*), the infant's C2 writes: "this event is dangerous"—the child cries.*
- *If the caregiver smiles (low* Γ^2*), the infant's C2 writes: "this event is harmless"—the child resumes play.*

This is not imitation: it is $\mathcal{S}_{\text{eff}}$ ***bootstrapping via external*** Γ^2 ***calibration**. The same mechanism operates in two directions:*

	Read other's Γ^2	*Effect*
Adult (empathy)	*Triggers matching Z-pattern*	Feels *the other*
Infant (referencing)	*Writes new Z-pattern into* $\mathcal{S}_{\text{eff}}$	Calibrates *futu*

This predicts the permanent impact of early caregiving: during the η_{peak} *period, Z-patterns are written at maximum depth. A caregiver who chronically transmits high* Γ^2 *(anxiety, hostility) calibrates the infant's amygdala baseline to "the world is dangerous"—a setting that persists in* $\mathcal{S}_{\text{eff}}$ *for life (cf. ACE studies; Felitti et al., 1998).*

Remark 8 (Therapeutic communication as guided Γ^2 descent)**.** *Chronic caregiver burden shares the* Γ*-topology of post-traumatic stress disorder (PTSD, Paper 4): a high-*Γ^2 *pattern on physical channels (*K_{sleep}*,* $K_{muscular}$*,* $K_{thermal}$*) that has been reinforced daily until it approaches* $\mathcal{S}_{\text{eff}}$*. Once near-fixed, the pattern cannot be displaced by language alone (*$K_{language}$ *does not reach the affected channels; Remark 6).*

Effective psychotherapy succeeds because the therapist provides structured x_{in} on the correct K-channels*, enabling a guided C2 descent:*

1. ***Resonance**: the therapist's* $\mathcal{S}_{\text{eff}}$ *contains matching Z-patterns (training + experience), so* $\Gamma^2_{empathy} \neq 0$*—the client* feels understood. *This opens* $G_{emotional}$*, which mere advice cannot.*

2. ***Guided re-exposure**: the therapist carefully represents* x_{in} *at a controlled intensity, allowing C2 to write a new* ΔZ *on the trauma channel without overwhelm—gradual gradient descent rather than a single large step that would overshoot.*

3. ***Action prescription**: effective therapy is not conversation alone but includes* behavioural homework*—physical action that provides x_{in} on somatic channels ($K_{muscular}$, $K_{thermal}$) where C2 can update.*

The therapeutic relationship is therefore a mutual C2 calibration*: the therapist and client jointly perform gradient descent on the client's Γ^2 landscape. The session provides η-elevation (safe novelty $\Rightarrow$ dopamine) and structured x_{in} on the right channels—precisely the two ingredients that unstructured social support cannot reliably supply.*

D. Reconciliation with established memory neuroscience

Remark 9 (The hippocampus as episodic C3 relay)**.** *Mainstream neuroscience identifies the hippocampus as the "memory centre" (Scoville & Milner, 1957). In the Γ-framework, the hippocampus is not a storage device but a* C3 relay specialised for episodic binding*: it provides the high-K cross-dimensional relay required to bind what/where/when/how into a single C2 tick.*

Patient H.M. (bilateral hippocampectomy) could not form new episodic memories but could learn new motor skills (Corkin, 2002). This is predicted by C2: removing the hippocampal C3 relay eliminates episodic binding (high-K cross-dimensional) but leaves single-channel C2 intact (motor learning in cerebellum, emotional conditioning in amygdala).

Memory type	*Needs hippocampus?*	*K*	*Γ-Net role*
Episodic	*Yes*	*High*	*C3 cross-dimensional relay*
Procedural	*No*	*Low*	*Cerebellar C2 (single-channel)*
Emotional	*No*	*Low*	*Amygdala direct C2*
Priming	*No*	*Low*	*Same-channel C2 pre-activation*

Memory consolidation during sleep (hippocampal replay $\rightarrow$ cortical transfer; Diekelmann & Born, 2010) is the C3 relay gradually "teaching" the cortex to reproduce the bound Z-pattern without the relay—transferring from hippocampal-dependent to hippocampal-independent $\mathcal{S}_{eff}$.

Remark 10 (Memory is the impedance configuration, not a stored object)**.** *The Γ-framework predicts that memory is not "stored in" the brain any more than a violin's timbre is "stored in" the violin. The Z-pattern* is *the memory: it is the configuration of every interface in the system. Recalling a memory is not retrieval from a separate storage medium but* re-activation *of the Z-pattern.*

Engram cells (Liu et al., 2012; Josselyn & Tonegawa, 2020) are real and experimentally verified: optogenetic stimulation of engram-labelled neurons elicits the associated memory. In Γ-Net terms, engram cells are entry nodes *of the Z-pattern: activating them provides the initial x_{in} that, via C3 relay, reconstructs the full pattern across all bound channels. They are not "containers" but "keys"—the Z-pattern they unlock is distributed across every interface that participated in the original binding.*

This interpretation originated from two clinical observations. First, peaceful elders retain complete autobiographical identity despite severe physical decline—if memory were a stored object it would degrade with the substrate, but it does not. Second, dementia patients lose memories in strict reverse chronological order (Remark 38)—the erasure sequence mirrors the η-weighted writing depth, exactly as predicted by C2. Together these observations establish that memory is the Z-pattern configuration*, with depth proportional to η at the time of writing.*

Remark 11 (LTP and LTD as two views of a single C2 update)**.** *Long-term potentiation (LTP) and long-term depression (LTD) are conventionally described as two separate calcium-dependent mechanisms (Malenka & Bear, 2004). In the Γ-framework they are a single C2 equation viewed from different channels.*

When C2 updates impedance toward a new pattern B:

$$Z \xrightarrow{\Delta Z} Z' \;:\quad \Gamma^2_B \downarrow \ (\textit{LTP for } B), \quad \Gamma^2_A \uparrow \ (\textit{LTD for } A)\,.$$

Moving Z toward the matching point of B necessarily moves it away from the matching point of any previously learned pattern A. LTP and LTD are therefore not mechanistically independent: they are the same impedance shift observed from two different input channels.

This predicts that LTP on one pathway and LTD on a competing pathway should be correlated in magnitude, a prediction consistent with heterosynaptic depression data (Chistiakova et al., 2014). More precisely, since both arise from a single ΔZ, the magnitudes satisfy

$$|\Delta\Gamma^2_{LTP}| + |\Delta\Gamma^2_{LTD}| = \textit{const} \quad (\textit{impedance conservation})\,,$$

whereas the Bienenstock–Cooper–Munro (BCM) model predicts a threshold switch *without constraining the magnitudes on competing pathways. A patch-clamp experiment measuring simultaneous LTP and LTD on two convergent pathways in CA1 could distinguish these predictions directly.*

Remark 12 (The cerebellum as a general-purpose low-K C2 co-processor)**.** *The cerebellum contains $\sim$80 % of the brain's neurons despite occupying only $\sim$10 % of its volume. In the Γ-framework, this reflects its function as a* massively parallel low-K C2 engine*. Although historically associated with motor control, the cerebellum also participates in emotional conditioning (Sacchetti et al., 2002) and cognitive timing (Strick et al., 2009). The Γ-framework explains this naturally: the Purkinje-cell architecture implements a general-purpose C2 loop—climbing-fibre error signal (Γ^2) drives parallel-fibre ΔZ—that operates on* any *low-K channel, not only motor channels.*

The vast neuron count is needed because real-time calibration requires simultaneous C2 updates on hundreds of channels in parallel, each at low K but with very fast

update rates ($\tau_{cerebellar} \approx 10\,ms$ vs. $\tau_{cortical} \approx 100\,ms$). Cerebellar damage produces ataxia *(loss of coordination, not loss of strength) because the C2 co-processor is offline: Γ^2 on motor channels cannot be minimised in real time, but the cortical C3 commands remain intact.*

XI. THE BINDING PROBLEM AS SYNCHRONOUS C2 WRITING

Multi-modal perception requires disparate sensory channels (vision, audition, touch, olfaction) to be unified into a single coherent percept. The classical "binding problem"—how the brain knows that a red apple is the same object as the sweet taste—has resisted solution because it presupposes a central integrator. In the Γ-framework, no such integrator is needed: *binding is synchronous C2 writing through shared C3 relay nodes.*

A. Binding by temporal coincidence

Each sensory modality α contributes an impedance change at its own modal complexity K_α: $\Delta Z_\alpha = -\eta\,\Gamma_\alpha\, x_{\text{in},\alpha}\, x_{\text{out},\alpha}$. Two channels α, β are *bound* if and only if their C2 updates are both non-zero within the same update tick τ:

$$\text{Bound}(\alpha,\beta,\tau) \iff \Delta Z_\alpha(\tau) \neq 0 \;\wedge\; \Delta Z_\beta(\tau) \neq 0\,. \quad (20)$$

The activity-gating factor $x_{\text{in}} \cdot x_{\text{out}}$ is the *coincidence detector*: only channels that are simultaneously active participate in the same write. The thalamus—the brain's principal C3 relay node—serves as the shared interface through which all cortical channels must pass, ensuring that co-active channels share the same C2 tick.

Remark 13 (Gamma oscillation as C2 clock)**.** *The 30–70 Hz gamma oscillation widely observed in cortical recordings corresponds, in this framework, to the C2 update clock rate. Each gamma cycle (∼25 ms) defines one binding window τ: channels whose $x_{in} \cdot x_{out} > 0$ within the same cycle are written together and thus bound. No additional "synchronisation mechanism" is required; C2 activity gating* is *the mechanism.*

B. Corpus callosum as inter-hemispheric C3 relay

The two cerebral hemispheres specialise in different K-ranges:

Left hemisphere	Right hemisphere
Sequential language ($K \sim O(10)$)	Spatial/holistic ($K \sim O(10^2)$)
Fine motor control	Global pattern recognition
Logical chains	Emotional colour

The corpus callosum ($\sim 2\times10^8$ axons, the largest white-matter tract) is the C3 K-staircase that matches the two hemispheres:

$$\text{L-hemisphere } (K_L) \underset{\text{corpus callosum}}{\longleftrightarrow} (K_R) \text{ R-hemisphere.} \quad (21)$$

Here K_L and K_R are order-of-magnitude estimates reflecting the sequential (~ 10 temporal features) versus spatial ($\sim 10^2$ pixel-like features) nature of the dominant processing mode in each hemisphere; precise values depend on the task-specific K-decomposition and are not required for the qualitative argument.

Callosotomy (split-brain surgery) severs this relay: $\Gamma_{L\leftrightarrow R} \to 1$, and the two hemispheres become independent networks, each with its own binding domain—precisely the split-brain phenomenology reported by Gazzaniga (1967).

Remark 14 (Lateralisation as impedance specialisation)**.** *Hemispheric lateralisation is not a design choice but a consequence of C3 efficiency. A single hemisphere covering both low-K sequential and high-K spatial modes would require a longer internal K-staircase with higher total Γ^2. Splitting into two specialised hemispheres with a dedicated inter-hemispheric relay (corpus callosum) minimises the system's total reflection—a direct prediction of Γ-MRP.*

Remark 15 (Hemispheric bifurcation as neural Murray's law)**.** *Murray's law for vascular branching states that a parent vessel bifurcates when two daughter vessels carry the same total flow at lower total impedance ($r_0^3 = r_1^3 + r_2^3$, Paper 2). The hemispheric split obeys the same Γ-MRP logic with neural impedance in place of hydraulic impedance. A single cortex spanning $K_{\min} \leftrightarrow K_{\max}$ requires an internal K-staircase of length $\log(K_{\max}/K_{\min})$ with cumulative reflection Γ^2_{single}. Splitting into two specialised hemispheres, each covering half the K-range, plus a matched inter-hemispheric relay of impedance $Z_{CC} = \sqrt{Z_L \cdot Z_R}$, yields:*

$$\Gamma_L^2 + \Gamma_R^2 + \Gamma_{CC}^2 \;<\; \Gamma^2_{single}\,, \quad (22)$$

which is the neural analogue of the vascular branching inequality. Both are instances of the same principle: Γ-MRP drives bifurcation whenever the total reflection of two matched sub-networks plus their relay is less than the reflection of a single unmatched network. *This unifies vascular topology (Paper 2) and neural topology under a single optimality condition.*

Remark 16 (Cortical area count as peripheral interface count)**.** *The HCP multi-modal parcellation identifies ∼360 cortical areas (180 per hemisphere; Glasser et al., 2016). The Γ-framework predicts that each distinct peripheral tissue interface requires a dedicated cortical C2 controller and C3 relay node. The human body contains ∼78 organs (anatomical consensus) distributed across ∼30 dermatomes (31 spinal nerve pairs*

minus C1), with bilateral symmetry and multi-modal innervation (motor, sensory, autonomic), yielding an estimated $78 \times 2 \times 2 \approx 312$ *independent interface channels—consistent with 360 cortical areas to within the counting uncertainty.*

The cortical magnification factor*—the disproportionately large cortical area devoted to fingertips, lips, and face (the sensory homunculus)—is conventionally attributed to peripheral innervation density alone. However, innervation density does not fully account for the observed magnification (Corniani & Saal, 2020). The* Γ*-framework provides the missing factor: cortical area* $\propto K_{peripheral}$*, the modal complexity of the tissue interface. Fingertips require simultaneous multi-modal binding (tactile texture + pressure + proprioception + temperature + fine motor control), yielding high* K *and therefore a large C3 relay area. The trunk, despite substantial physical area, has low multi-modal binding demands (low* K*) and correspondingly small cortical representation.*

This somatotopic organisation is present even in preterm infants (Dall'Orso et al., 2018), consistent with Z_0^{DNA} *(genetically fixed impedance, Theorem 5): the cortical map is a boundary condition, not a learned structure.*

Remark 17 (Tree-trunk topology of the central nervous system)**.** *The nervous system's architecture maps onto a tree:*

Tree	*CNS*	Γ*-Net role*
Canopy (leaves)	*Cerebral cortex*	*High-K sensory/cognitive processing*
Trunk	*Spinal cord*	*C3 main relay (K-staircase backbone)*
Root system	*Peripheral nerves → organs*	*Low-K end-effectors (C2 update targets)*

Three quantitative predictions follow from this topology:

1. ***Trunk diameter obeys Murray's law.*** *The spinal cord is thickest at the cervical enlargement (carrying all limb and organ channels), thins in the thoracic region (upper-limb signals already diverted), and widens again at the lumbar enlargement (lower-limb signals converge). Cross-sectional area at each level equals the Murray-optimal sum of all downstream branch areas.*

2. ***Trunk severance = downstream*** $\Gamma \to 1$***.*** *Spinal cord injury at level* ℓ *sets* $\Gamma_k = 1$ *for all channels* $k > \ell$ *(below the injury): complete reflection, zero transmission, paralysis. Channels* $k < \ell$ *(above the injury) remain intact. This is exactly the clinical presentation of spinal cord injury, derived here from a single equation.*

3. ***Bidirectional flow, like sap.*** *Energy flows upward (sensory* x_{in} *from organs to cortex) and downward (motor* x_{out} *from cortex to organs) simultaneously, with the spinal cord providing C3 impedance matching in both directions. Damage to ascending tracts (posterior columns) produces sensory loss without motor deficit; damage to descending tracts (corticospinal) produces motor deficit without sensory loss—the two flows are physically independent channels sharing the same trunk.*

4. ***Reflex arcs as spinal-level C3 shortcuts.*** *The full sensory–cortical–motor loop traverses the entire* K*-staircase twice (ascending + descending), accumulating* Γ^2 *at every relay. For survival-critical responses,* Γ*-MRP selects a shorter path: the* spinal reflex arc *closes the C2 loop at the spinal level, bypassing cortical processing entirely.*

Path	*C3 stages*	*Latency*	*Example*
Reflex arc	*∼2*	*∼30 ms*	*Knee jerk*
Cortical loop	*∼14*	*∼300 ms*	*P300 ERP*

The ∼7× stage-count ratio (14/2) compares with the observed ∼10× latency ratio (300/30). The discrepancy is expected: synaptic delay is not perfectly uniform across relay types (chemical synapses in cortex ≈ 1–2 ms each vs. electrical gap junctions in some spinal interneurons ≈ 0.1 ms), and the cortical loop includes additional thalamocortical relays not counted as C3 stages. The prediction is that the ratio should lie in the range $[N_{reflex}/N_{cortical},\ \tau_{cortical}/\tau_{reflex}]$*, bracketed by stage count and total latency. The subjective experience of "withdrawing before feeling pain" is a direct consequence: the motor reflex completes in ∼2 C3 stages while pain perception requires the full ∼14-stage ascent to cortex.*

C. Sensory misalignment and compensatory behaviour

When two bound channels arrive out of synchrony—for example, seeing lip movement at tick t but hearing the voice at tick $t + \Delta$—the binding condition (Eq. 20) fails. The organism's learned Z-pattern expects synchrony; the mismatch elevates Γ^2, triggering the Perception–Error–Compensation loop (Paper 0, Perception–Error–Compensation Remark):

Δ range	Compensation	Percept
< 150 ms	C2 temporal recalibration	"Normal" (unnoticed)
150–300 ms	Head turn, gaze shift	"Something is off"
> 300 ms	Binding abandoned	Two separate events

The ventriloquism effect—sound perceived at the visual source location—is C2 overwriting the auditory spatial map with the dominant visual Z-pattern, because vision has higher spatial K and therefore lower Γ^2 on the spatial dimension.

D. Missing-channel compensation

Remark 18 (Cross-modal plasticity as C2 repurposing). *When a sensory channel is permanently absent ($x_{in,\alpha} = 0$ $\forall t$), C2 produces $\Delta Z_\alpha = 0$: the cortical territory assigned to channel α is never written. However, neighbouring channels with $x_{in} > 0$ provide lateral x_{out} that invades the unused territory. C2 then writes* their *Z-patterns into the vacated substrate:*

- ***Congenital blindness**: visual cortex V1 is rewritten by tactile and auditory inputs (Braille reading activates V1; Sadato et al., 1996).*
- ***Congenital deafness**: auditory cortex A1 is rewritten by visual and somatosensory inputs (sign-language processing in A1; Nishimura et al., 1999).*

No special "plasticity mechanism" is required; C2 with activity gating automatically repurposes unused Z-substrate.

Remark 19 (Deafness and language: the C3 re-routing prediction). *Language is a Z-pattern, not a sound. In hearing individuals, the language K-staircase runs through the auditory channel. In congenital deafness, $x_{in,auditory} = 0$, so C2 never writes the auditory language pathway. Sign language re-routes the same Z-pattern through the visual–motor channel: the C2 equation is substrate-independent (Paper 3, C2 universality). If sign language is introduced after the developmental η-peak (age $\sim$5), the visual language K-staircase is incompletely formed—predicting the well-documented "language deprivation syndrome" in late-signing deaf children.*

E. Self-talk as C3 dimensional compression

Remark 20 (Verbalisation as external C3 loop). *Internal cognitive states reside in the PFC at modal complexity $K_{PFC} \sim O(10^2)$. When internal Γ^2 is high (emotional activation, confusion, conflict), Γ-MRP drives the system to find a discharge pathway. The language channel provides a well-trained K-staircase:*

$$K_{PFC}\,(\sim 10^2) \xrightarrow[Broca]{C3} K_{language}\,(\sim 10^1) \xrightarrow[motor]{C3} K_{voice}\,(\sim 10^0) \xrightarrow[Wernicke]{C3} K_{PFC}$$

The compression from $K \sim 10^2$ to $K \sim 10^1$ forces the speaker to discard irrelevant dimensions, retaining only the most salient features of the internal state. This is why "saying it out loud" clarifies thinking—the C3 compression is a physical filter. Writing provides the same compression with an additional advantage: the output persists as a visual x_{in} that can be re-read, enabling iterative C2 refinement.

Remark 21 (Language fluency vs. content novelty). *The language* channel *(phonology, grammar) converges to $\Gamma^2 \approx 0$ after years of C2 update and enters $\mathcal{S}_{eff}$. Fixed phrases ("hello," "thank you") are fully compressed paths with $\Gamma^2 \approx 0$. However, sentences are* compositional*: each novel combination of words requires a new $K_{PFC} \to K_{language}$ compression mapping that has never been executed. The subjective experience of "thinking about what to say" is therefore not language processing (which is automatic) but real-time C3 compression of a novel high-K state into the low-K language channel.*

Remark 22 (Environmental noise as binding dilution). *C2 activity gating does not distinguish signal from noise: any channel with $x_{in} \cdot x_{out} > 0$ participates in the current write. In a quiet environment, only the target channels are active; C2 writes a clean, high-contrast Z-pattern. In a noisy environment, many irrelevant channels are simultaneously activated by ambient stimuli, diluting the binding:*

$$SNR_{binding} = \frac{\sum_{target} |\Delta Z_\alpha|}{\sum_{noise} |\Delta Z_\beta|} < \theta_{recall} \Rightarrow \text{memory retrieval fails.}$$

This provides the physical basis for two well-known observations:

1. ***Learning requires low-noise environments**: effective C2 binding demands that target-channel $|\Delta Z|$ dominates the total write. Classrooms, libraries, and study environments minimise $\sum_{noise} |\Delta Z_\beta|$.*
2. ***Context-dependent memory**: environmental noise that* was *present during learning is written into the Z-pattern alongside the target. Recall is facilitated when the same noise is re-presented (reinstating the bound x_{in} pattern), consistent with encoding-specificity findings (Tulving & Thomson, 1973).*

Remark 23 (Metabolic modal transfer and subjective relaxation). *In a familiar environment, most impedance mismatches are same-dimensional: the learned Z-pattern is well-matched ($\Gamma^2 \approx 0$), C2 updates are small, and C3 is rarely triggered. The brain operates in "autopilot"—metabolic cost is dominated by low-K routine channels. Over time, un-updated channels accumulate impedance debt D_Z while the novelty-dependent learning rate η declines (no new x_{in} patterns). The subjective experience is stale-mind and fatigue.*

Entering a radically different environment (e.g. international travel) simultaneously invalidates Z-patterns across multiple K-dimensions: language (K_{ling}), food ($K_{gustatory}$), visual scenery (K_{visual}), social norms (K_{social}), and climate ($K_{thermal}$). The massive cross-dimensional mismatch triggers widespread C3 relay activation, forcing C2 to recalibrate many channels simultaneously. The metabolic cost shifts from low-K routine channels to high-K novelty channels:

$$\frac{d}{dt}\left(\frac{\Sigma\Gamma^2_{high\text{-}K}}{\Sigma\Gamma^2_{total}}\right) > 0\,.$$

Crucially, novelty elevates dopaminergic η (Paper 3, Lifecycle Equation), re-enabling C2 updates on channels that had been stagnant. The subjective experience is "tiring but refreshing" —high metabolic expenditure, but directed at productive ΔZ rather than chronic D_Z accumulation.

The prediction extends to return-from-travel syndrome: upon return, the familiar environment re-invalidates the travel-updated Z-patterns while novelty-driven η has already subsided, producing high Γ^2 with low η—the precise condition for subjective malaise.

Remark 24 (Dopamine as the mathematically necessary η-modulator)**.** *A fixed learning rate η is unstable: too high causes oscillation, too low prevents adaptation. Γ-MRP requires η to be dynamic. Variational optimisation of the reflection action $\mathcal{A}[\Gamma] = \int \Sigma\Gamma^2\, dt$ with respect to η yields:*

$$\eta^*(t) \;\propto\; -\frac{d\,\Gamma^2}{dt}\,. \tag{23}$$

The optimal η is large *when Γ^2 is decreasing ("correct direction—accelerate learning") and* small *when Γ^2 is increasing or stationary ("wrong direction—stop"). This is precisely the function of the dopaminergic system: ventral tegmental area (VTA) neurons fire when outcome exceeds prediction (negative reward prediction error $\equiv d\Gamma^2/dt < 0$), releasing dopamine that amplifies synaptic plasticity (long-term potentiation). Dopamine is therefore not a "reward molecule" but an η-gain signal—a mathematical necessity of C2, not an evolutionary design choice.*

Addiction *follows directly: exogenous substances (nicotine, opioids, amphetamines) hijack the η-modulator, artificially setting $\eta \gg \eta^*$ on the reward channel. The Z-pattern of substance use is written into $\mathcal{S}_{\text{eff}}$ with far fewer repetitions than normal motor learning ($\eta \times n = const$; high η compensates for low n). Once inside $\mathcal{S}_{\text{eff}}$, the pattern is irremovable (Theorem 7), explaining lifelong relapse vulnerability.*

Remark 25 (The neural operating cycle)**.** *The binding, learning, and recovery mechanisms of this section form a single closed operating cycle:*

1. ***Wake: C2 + dopamine.*** *Γ-MRP drives C2 updates; dopamine modulates η via Eq. (23).*

2. ***Metabolic cost.*** *Each $\Delta Z \neq 0$ consumes ATP. ATP hydrolysis yields ADP $\rightarrow$ AMP $\rightarrow$* adenosine*, which accumulates monotonically during wakefulness.*

3. ***Shutdown trigger.*** *Adenosine binds A_1 receptors on wake-promoting neurons, progressively inhibiting arousal ($Z_{arousal}\downarrow$) until the Television Inequality (Theorem 4) fails: $Z_{arousal} \cdot \sum G_i \cdot Q_{PFC} < \theta \Rightarrow$ loss of consciousness (sleep).*

4. ***Sleep: clearance + reset.*** *Glymphatic flow clears adenosine and metabolic waste (D_Z reduction). Gate-leak processing during REM (Paper 4) replays and consolidates the day's Z-updates.*

5. ***Wake: cycle restarts.*** *Adenosine cleared $\rightarrow$ $Z_{arousal}$ restored $\rightarrow$ Television Inequality satisfied $\rightarrow$ consciousness returns.*

This five-step cycle is not a biological design but a physical consequence *of C2: any system that updates impedance by gradient descent must consume energy, accumulate waste, and periodically halt to clear that waste. The infant's 16-hour sleep schedule and the adult's 8-hour schedule differ only in η: higher η (developmental peak) $\rightarrow$ faster ATP consumption $\rightarrow$ faster adenosine accumulation $\rightarrow$ more frequent and longer sleep.*

Remark 26 (Dreaming as gate-leak binding)**.** *During REM sleep, thalamic gates are partially open (gate leak, $0 < G_i \ll 1$; Paper 4, § Sleep disorders). In this regime, C2 still executes $\Delta Z = -\eta\,\Gamma\, x_{in}\, x_{out}$ but the x_{in} vectors are internally generated (no external stimulation), drawn from the day's residual activation patterns. The partial-gate condition admits multiple channels simultaneously but without the structured x_{in} of wakefulness. The binding condition (Eq. 20) is satisfied by whichever channels happen to have residual $x_{in} \cdot x_{out} > 0$ in the same tick—a stochastic process. The result is* dreams*: coherent but unexpected bindings of memory fragments that would never co-occur during structured wakefulness. The "narrative logic" of dreams reflects the C3 relay topology (channels that share a relay node are more likely to be co-activated), while the "bizarreness" reflects the random gating (channels that normally never bind are forced together by chance).*

Remark 27 (Attention as gate selection; ADHD as gate instability)**.** *Selective attention corresponds to setting one thalamic gate high ($G_{target} \approx 1$) while suppressing all others ($G_{j\neq target} \approx 0$). This maximises the binding SNR (Remark 22) by ensuring that only target-channel ΔZ dominates the write. Each gate G_i has an RC time constant τ_G determined by the prefrontal–thalamic feedback loop. Sustained attention requires τ_G to exceed the task duration: the gate must remain open long enough for C2 to complete meaningful Z-updates.*

ADHD is, in this framework, a shortened τ_G:

$$\tau_{G,ADHD} \ll \tau_{G,typical} \Rightarrow \text{gate switches before C2 completes.} \tag{24}$$

The consequence is not inability to learn (C2 is intact) but inability to bind *a coherent Z-pattern before the gate switches to another channel. Stimulant medications (methylphenidate, amphetamine) increase dopaminergic tone in the PFC, effectively lengthening τ_G by stabilising the prefrontal–thalamic loop—consistent with the η-modulation framework of Remark 24.*

Remark 28 (Emotion as low-K broadcast)**.** *The amygdala receives sensory input via a* subcortical shortcut *(thalamus → amygdala, bypassing cortex; LeDoux, 1996). This pathway has very low K ($\sim O(1)$): it carries only coarse threat/reward classification, not detailed feature analysis. In Γ-Net terms, the amygdala is a* low-K fast C1 detector*: it evaluates Γ^2 on basic survival channels (predator, food, mate) before the high-K cortical analysis is complete.*

When $\Gamma^2_{amygdala}$ exceeds a threshold, the amygdala triggers a system-wide response:

- ***Autonomic broadcast**: sympathetic activation changes $Z_{cardiac}$, $Z_{vascular}$, $Z_{respiratory}$ simultaneously (fight-or-flight).*
- ***Gate override**: $G_{threat} \to 1$, all other $G_i \to 0$ (attentional capture by emotional stimulus).*
- ***η amplification**: amygdala → VTA projection releases dopamine, ensuring the threatening Z-pattern is written with high η (one-trial fear learning).*

Emotional "irrationality" is therefore not a failure of the system but the intended low-K fast path overriding the slow high-K cortical analysis—a Γ-MRP optimal strategy when speed matters more than precision.

Remark 29 (Creativity as gate relaxation)**.** *Creativity requires binding channels that are normally separated—connecting ideas from different K-domains that would not co-occur in focused attention. In the framework, this corresponds to* relaxing *the gate selectivity: instead of $G_{target} = 1$, $G_{other} = 0$ (focused attention), the creative state sets $G_i \approx 0.3$ across many channels simultaneously.*

This increases the number of channels that pass the binding condition (Eq. 20) per tick, enabling remote associations*—bindings between channels with large K-separation that are normally isolated. The cost is reduced binding SNR (Remark 22): many weak bindings instead of one strong binding.*

This explains two observations:

1. ***Creativity and relaxation co-occur**: showers, walks, and drowsiness are creative because prefrontal gate control weakens, allowing wider binding.*
2. ***Creativity and focus are antagonistic**: tight gate ($G_{target} = 1$) blocks remote associations; loose gate allows them but prevents deep single-channel processing. The "incubation effect" is the alternation between these two modes.*

Remark 30 (Intuition as $\mathcal{S}_{\text{eff}}$ direct output)**.** *Expert intuition—the immediate "knowing" without conscious reasoning—is the output of a Z-pattern that has entered $\mathcal{S}_{eff}$ through thousands of C2 updates. The pattern is so deeply written ($\Gamma^2 \approx 0$) that it does not require PFC processing: the input x_{in} directly produces x_{out} through the fixed Z, bypassing the high-K cortical reasoning loop entirely.*

In contrast, novice reasoning must traverse the full K-staircase (input → PFC analysis → decision → output), accumulating C3 relay delays and Γ^2 at each stage. The subjective difference between "I know this is right" (intuition) and "let me think about it" (reasoning) corresponds to:

	Intuition	*Reasoning*
Z-pattern location	$\mathcal{S}_{eff}$	*Active Z (still updatin*
C3 stages traversed	*$\sim$2 (reflex-like)*	*$\sim$14 (full cortical loop*
Latency	*$\sim$100 ms*	*$\sim$1–10 s*
Confidence	*High (but not infallible)*	*Variable*

Intuition can be wrong when the environment changes but $\mathcal{S}_{eff}$ has not—the pattern is frozen and cannot update. This is the expert's "experience trap": the same mechanism that enables instant correct responses in familiar situations produces instant wrong *responses in unfamiliar situations that superficially resemble familiar ones.*

Remark 31 ("Sixth sense" as sub-threshold amygdala detection)**.** *The subjective experience of "sensing danger without knowing why" is not a sixth sensory modality but the output of the amygdala's low-K fast pathway (Remark 28) processing sub-threshold inputs from the standard five senses. Peripheral cues—a subtle change in ambient sound, a faint odour, a movement in peripheral vision—enter the amygdala via the subcortical shortcut in 30–74 ms (Remark 3), triggering $\Gamma^2_{threat} > 0$ before PFC has completed conscious evaluation (200–300 ms).*

The subjective sequence is:

1. *"Something feels wrong" (amygdala output, no PFC involvement).*
2. *"I don't know why" (PFC has not yet processed the cue).*
3. *Seconds later: "Ah,* that's *what it was" (PFC completes analysis and identifies the cue).*

The "sixth sense" is the temporal gap between amygdala detection and cortical awareness, not evidence of an additional sensory channel.

Remark 32 (Remote sensing: physical limits and the $\mathcal{S}_{\text{eff}}$ alternative)**.** *Reports of "sensing a loved one in danger at a distance" are physically constrained by the signal-to-noise ratio of brain-to-brain electromagnetic coupling. The brain emits $\sim 10^{-12}$ W of electromagnetic power; the Johnson–Nyquist noise floor at the receiver is $P_{noise} = k_B T \Delta f \approx 4 \times 10^{-17}$ W. At the scalp, SNR $\approx 10^5$ (EEG works), but power falls as $1/r^2$; beyond $\sim$3 cm, SNR < 1 and no information can be decoded.* Brain-to-brain electromagnetic communication at macroscopic distances is therefore physically impossible.

The Γ-framework offers a non-supernatural alternative:

1. $\mathcal{S}_{\mathit{eff}}$ ***convergence***: *years of co-habitation produce mutual C2 calibration, making the two S_{eff} nearly identical. Both individuals respond to the same environmental risk cues with the same Γ^2 pattern.*

2. ***Sub-threshold cue storage***: *during the last face-to-face interaction, the partner's micro-expressions (vocal pitch, gaze aversion, respiration rate) were detected by the observer's amygdala and written into a transient Z-buffer. Background processing of this buffer may complete hours later, producing anxiety that coincides with the actual event.*

3. ***Confirmation bias***: *worry episodes are frequent (∼100/year); actual events are rare (∼1/year). Only coincidences are remembered (high-η emotional write), inflating subjective accuracy to ∼100 % against a base rate of ∼1 %.*

Remark 33 (Self-confidence as pre-PFC action threshold)**.** *Self-confidence is the willingness to act on $\mathcal{S}_{\mathit{eff}}$ output (amygdala/intuition, ∼30–100 ms) before PFC confirmation (∼200–300 ms). Whether this willingness exists depends on a single Z-pattern: "my sub-threshold Γ^2 detections are reliable."*

	High confidence	*Low confidence*
Z-pattern in S_{eff}	*"trust intuition"*	*"distrust intuition"*
Action pathway	*Amygdala → direct output*	*Amygdala → wait for PFC*
Response latency	*∼100 ms*	*∼1–10 s*

This pattern is trained by C2: each time a sub-threshold detection is validated *by subsequent events, dopamine (η ↑) reinforces the "trust" Z-pattern (LTP). Each time it is* invalidated *or* dismissed by others, *the competing "distrust" pattern is reinforced (LTD for "trust"). Chronic invalidation (controlling relationships, repeated failure) writes "distrust" deeply into $\mathcal{S}_{\mathit{eff}}$, producing low self-confidence that is resistant to verbal encouragement ($K_{language}$ cannot reach the affected somatic channels; Remark 8). Rebuilding confidence requires structured* experiential *success—providing x_{in} on the correct channels to drive C2 back toward the "trust" Z-pattern.*

XII. SOUL AS THE NULL SPACE OF LEARNING

A. What cannot be changed by experience

Definition 3 (Soul (Invariant Topological Core))**.**

$$\mathcal{S} = \ker\left(\frac{\partial Z}{\partial t}\right) = \{Z_k : \Delta Z_k = 0 \ \ \forall \ \mathit{experience}\}\,. \quad (25)$$

Theorem 5 (Soul Decomposition)**.**

$$\mathcal{S} = A_{cut} + Z_0^{DNA} + \rho^2_{irred}\,, \quad (26)$$

where: (1) A_{cut}: the irreducible topological action cost (Theorem 1); (2) Z_0^{DNA}: genetically fixed characteristic impedance, not modifiable by C2; (3) ρ^2_{irred}: the irreducible noise floor from Johnson–Nyquist thermal physics.

Proof. (i) $A_{\rm cut}$ is invariant by Theorem 1. (ii) C2 modifies $Z_{L,k}$, not $Z_{0,k}$; hence $Z_0^{\rm DNA}$ is invariant. (iii) $\rho^2_{\rm irred}$ is a thermodynamic constant. □

Corollary 4 (The soul is not an event)**.** *$\mathcal{S}$ does not "arrive" at a moment or "depart" at death. It is the boundary condition of the impedance network.*

Corollary 5 (Uniqueness)**.** *A_{cut} depends on the specific topology (K_i, K_j), determined by genetics and stochastic developmental morphogenesis. No two organisms have identical A_{cut}.*

Remark 34 (Developmental genesis of the null space)**.** *Early-acquired patterns—native language, core motor programs, temperamental reflexes—become* functionally immutable *as $\eta(t) \to 0$ (the Arrhenius decline of Paper 3). They drift into an* effective null space $\mathcal{S}_{\mathit{eff}}(t) \supseteq \mathcal{S}$ *that grows with age.*

Remark 35 (Fixation as impedance physics)**.** *A pattern written during the η-peak and reinforced for decades has migrated into $\mathcal{S}_{\mathit{eff}}$—it is inside the null space. "I know this is irrational, but I cannot change it" is veridical: the speaker PFC correctly diagnosed that $\eta\,\Gamma\,x_{in}\,x_{out} \approx 0$ on the relevant sub-network.*

Remark 36 (Moral blame as secondary Γ^2 injection)**.** *Blame provides neither the sustained gradient nor an increase in η. The net effect is $\Delta(\Sigma\Gamma^2) > 0$ with $\Delta Z_{corrective} \approx 0$: load is added, but no repair path is opened—the precise opposite of therapy.*

Remark 37 (Reincarnation as null-space isomorphism)**.** *If past-life memory reports reflect null-space isomorphism, then:*

$$\mathcal{S}_A \cong \mathcal{S}_B \quad \Longleftrightarrow \quad \left(A^{(A)}_{cut}, Z_0^{DNA,(A)}, \rho^{2,(A)}_{irred}\right) \approx \left(A^{(B)}_{cut}, Z_0^{DNA,(B)}, \rho^{2,(}_{irr}\right. \quad (27)$$

This is not a metaphysical claim; it is a falsifiable prediction (Appendix B).

B. Mortality of the null space

Define the basal reflection:

$$\Gamma^2_{\rm basal}(t) = A_{\rm cut}(t) + \rho^2_{\rm irred}(t)\,. \quad (28)$$

By C1, the *death condition* is:

$$\boxed{\Gamma^2_{\rm basal}(t_*) \geq 1 \quad \Longrightarrow \quad T(t_*) \leq 0\,.} \quad (29)$$

When the minimum reflection cost exceeds the available transmission capacity, no signal can propagate. The soul is unteachable, but it is not immortal.

The unlearnable debt accumulates monotonically:

$$D_Z^{\text{irred}}(t) = \int_0^t \Gamma^2_{\text{basal}}(\tau)\, d\tau\,, \qquad \frac{dD_Z^{\text{irred}}}{dt} > 0 \;\; \forall\, t\,. \quad (30)$$

C. Four laws of the null space

Theorem 6 (Zeroth law: null-space equivalence)**.** *$\mathcal{S}_A = \mathcal{S}_B$ iff they share A_{cut}, Z_0^{DNA}, and ρ^2_{irred}.*

Proof. By definition, $\mathcal{S} = \ker(\partial Z/\partial t)$ is the set of impedance channels with zero C2 gradient. The gradient $\partial Z_k/\partial t = -\eta_k \Gamma_k x_{\text{in},k} x_{\text{out},k}$ vanishes for channel k under three conditions: $\eta_k = 0$ (genetically fixed substrate, encoded in Z_0^{DNA}); $\Gamma_k = 0$ (perfectly matched, contributing to A_{cut}); or $x_{\text{in},k} \cdot x_{\text{out},k} = 0$ (never co-activated, contributing to ρ^2_{irred}). Two systems share $\mathcal{S}$ iff all three components match. □ □

Theorem 7 (First law: conservation under C2)**.** *$\Delta_{\text{C2}}\,\mathcal{S} = 0$. No experience can alter $\mathcal{S}$.*

Proof. For any channel $k \in \mathcal{S}$, $\partial Z_k/\partial t = 0$ by definition. The C2 update $\Delta Z_k = -\eta_k \Gamma_k x_{\text{in},k} x_{\text{out},k}$ equals zero for all $k \in \mathcal{S}$ for the same reasons established in the Zeroth Law proof. Therefore $\mathcal{S}$ is invariant under C2. □ □

Theorem 8 (Second law: monotone expansion)**.**

$$\frac{d\,\dim\,\mathcal{S}_{\text{eff}}(t)}{dt} \;\geq\; 0 \qquad \forall\, t\,. \quad (31)$$

The effective null space only expands; it never contracts.

Proof. Since $\eta(t) = \eta_0\, f_{\text{dev}}(t)\, e^{-E_a/k_B T_{\text{age}}}$ is monotonically decreasing after the developmental peak, any channel that enters $\mathcal{S}_{\text{eff}}$ remains in it. □

Corollary 6 (Ribot's gradient as the Second Law)**.** *Ribot's Law (1881)—old memories resist neurodegeneration longer than recent ones—is the clinical shadow of Eq. (31).*

Theorem 9 (Third law: asymptotic completeness)**.**

$$\lim_{t\to\infty} \frac{\dim\,\mathcal{S}_{\text{eff}}(t)}{\dim\,Z_{\text{total}}} \;=\; 1\,. \quad (32)$$

Proof. By the Second Law (Theorem 8), $\dim\,\mathcal{S}_{\text{eff}}(t)$ is monotonically non-decreasing. The update rate $\eta(t) = \eta_0 f_{\text{dev}}(t) e^{-E_a/k_B T_{\text{age}}}$ decreases monotonically after the developmental peak (Arrhenius aging, Paper 3). For any channel $k \notin \mathcal{S}_{\text{eff}}(t)$, the time to enter $\mathcal{S}_{\text{eff}}$ is $\tau_k \sim 1/\eta(t)$, which diverges as $\eta \to 0$. In the limit $t \to \infty$, $\eta \to 0$, so all channels satisfy $\partial Z_k/\partial t \to 0$, meaning $\dim\,\mathcal{S}_{\text{eff}} \to \dim\,Z_{\text{total}}$. □ □

Corollary 7 (The consciousness–soul duality)**.** *As $\mathcal{S}_{\text{eff}} \to Z_{\text{total}}$, the substrate for consciousness vanishes: $R_{\text{seal}} \to 1 \Rightarrow \Phi_{\text{meta}} \to 0$. A perfect soul and complete consciousness are mutually exclusive.*

Theorem 10 (Death as mathematical necessity)**.** *Given the Second Law and $\Gamma^2_{\text{basal}} \leq 1$ from C1, there exists a finite t^* such that $\Gamma^2_{\text{basal}}(t^*) = 1$ and $T(t^*) = 0$.*

Proof. By the Second Law, $d\Gamma^2_{\text{basal}}/dt > 0$. By C1, $\Gamma^2 + T = 1$, so Γ^2_{basal} is bounded above by 1. A bounded, monotonically increasing function reaches the bound in finite time. □

Remark 38 (Peaceful ageing vs. dementia: two modes of $\mathcal{S}_{\text{eff}}$ at end of life)**.** *The death theorem predicts that every individual reaches $\Gamma^2 \to 1$, but it does not specify* how *the null space erodes. Observation of two end-of-life trajectories motivates the following distinction.*

Peaceful ageing. *η decays uniformly across all channels (Second Law). No new patterns enter $\mathcal{S}_{eff}$, but none are removed either: the elder retains a complete identity—knowing who they are, who they love, and what they have lived—until the body's Γ^2_{basal} reaches 1. $\mathcal{S}_{eff}$ is* intact at the moment of death.

Dementia (Alzheimer's type). *η drops* asymmetrically*: specific cortical regions (hippocampus, entorhinal cortex) lose C2 capacity first, while primary sensory and motor areas are preserved. $\mathcal{S}_{eff}$ erodes in reverse chronological order:*

Lost first	*Lost last*	*Γ-Net reason*
Recent memories	*Childhood memories*	*Low-η writes are s*
Names of acquaintances	*Mother's face*	*High-η_{peak} writes a*
Executive planning	*Motor habits (walking)*	*High-K PFC fails*

The subjective horror *of dementia has a precise physical description: $\mathcal{S}_{eff}$ disintegrates before $\Gamma^2_{basal} \to 1$. The "soul" departs while the body remains alive.*

This distinction arose from observing that peaceful elders retain complete autobiographical identity despite severe physical decline—implying that $\mathcal{S}_{eff}$ integrity is independent of Γ^2_{basal}. The reverse question—"what if $\mathcal{S}_{eff}$ erodes first?"—yields the dementia phenotype as a prediction*, not a post-hoc label.*

D. The Moral Constraint Theorem

Theorem 11 (Moral Constraint Theorem)**.** *Let a moral demand target an impedance pattern Z_k. The net physical effect is:*

$$\Delta\Gamma^2_{\text{net}} = \begin{cases} -\Delta\Gamma^2 & \text{if } Z_k \notin \mathcal{S}_{\text{eff}} \text{ and a C2 gradient is supplied,} \\ +\Gamma^2_{\text{blame}} & \text{if } Z_k \in \mathcal{S}_{\text{eff}} \text{ (no C2 path exists).} \end{cases} \quad (33)$$

The second case is physically indistinguishable from injury.

Proof. **Case 1**: $Z_k \notin \mathcal{S}_{\text{eff}}$. A C2 gradient exists: $\partial Z_k/\partial t = -\eta_k \Gamma_k x_{\text{in},k} x_{\text{out},k} \neq 0$. A moral demand that supplies the appropriate activity pattern $(x_{\text{in},k}, x_{\text{out},k})$

therefore drives Γ_k^2 toward zero, reducing mismatch. The net effect is $\Delta\Gamma^2_{\text{net}} = -\Delta\Gamma^2 < 0$.

Case 2: $Z_k \in \mathcal{S}_{\text{eff}}$. By the First Law (Theorem 7), $\Delta_{\text{C2}}\mathcal{S} = 0$: no C2 gradient can alter Z_k. The moral demand cannot produce the intended impedance change. However, the demand itself constitutes an input signal that creates a mismatch at the interface between the moral agent and the target system. This mismatch generates reflected power $\Gamma^2_{\text{blame}} P_{\text{in}} > 0$, which is absorbed by the target as thermal dissipation— physically identical to an injury signal. Therefore $\Delta\Gamma^2_{\text{net}} = +\Gamma^2_{\text{blame}} > 0$. □ □

Corollary 8 (Boundary of moral responsibility)**.** *Moral responsibility is physically meaningful only for patterns outside* $\mathcal{S}_{\text{eff}}$.

Part D: Experimental Evidence and Extensions

XIII. THE TELEVISION THEOREM

Theorem 12 (Television Theorem)**.** *Any system with three multiplicative components—a power supply P, a channel selector G, and a processor Q—produces an output $O = P \cdot G \cdot Q$. The system is conscious if and only if there exists a feedback path $\pi : O \to$ input such that Γ_{meta} is defined and measurable.*

Proof. **Necessity**: For $\Phi_{\text{meta}} > 0$, the meta-reflection coefficient $\Gamma_{\text{meta}} = (Z_{\text{obs}} - Z_{\text{act}})/(Z_{\text{obs}} + Z_{\text{act}})$ must be defined. This requires Z_{obs} and Z_{act} to both be measurable, which in turn requires a feedback path $\pi : O \to$ input that routes the output back to become the observer's input at the next tick. Without π, Z_{obs} is undefined and Γ_{meta} cannot be computed.

Sufficiency: Given $P > 0$, $G > 0$, $Q > 0$, and a feedback path π, the output $O = P \cdot G \cdot Q > 0$ is routed back as input. The observer sub-network samples O and produces Z_{obs}; the actor sub-network produced Z_{act} at the previous tick. Γ_{meta} is now well-defined (both impedances are finite and positive), and the Television Inequality (Theorem 4) determines whether $\Phi_{\text{meta}} > 0$. Therefore consciousness requires and is supported by the feedback path π. □ □

XIV. THERMAL VULNERABILITY AND THE K-DIMENSIONAL BOTTLENECK

Johnson–Nyquist thermal noise generates:

$$V_n^2 = 4k_B T \cdot R \cdot \Delta f \,. \tag{34}$$

TABLE III. Three systems with identical multiplicative structure but different feedback topology.

	Rock	Television	Brain
Power P	None	Wall plug	Brainstem
Gate G	None	Tuner	Thalamus
Processor Q	None	Video IC	PFC
$P{\cdot}G{\cdot}Q > 0$	No	**Yes**	**Yes**
Feedback π	No	No	**Yes**
Γ_{meta} defined	No	No	**Yes**
Conscious?	No	No	**Yes**

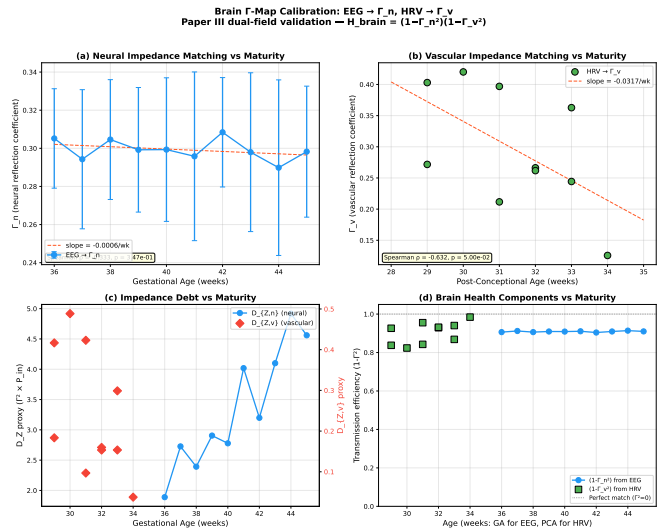

FIG. 6. Brain Γ-calibration across cortical regions. Higher-K regions show steeper thermal vulnerability. Simulated via `exp_brain_gamma_calibration.py`.

For a K-dimensional impedance matrix, the fractional perturbation scales as:

$$\frac{\delta\Gamma}{\Gamma} \propto \kappa(\mathbf{Z}) \cdot \frac{\delta T}{T} \propto K^{3/2} \cdot \frac{\delta T}{T} \,. \tag{35}$$

TABLE IV. Thermal vulnerability by axon dimension K. ($\delta T = +2\,\text{K}$, $T_0 = 310\,\text{K}$).

K	$\kappa(\mathbf{Z})$	$\delta\Gamma/\Gamma$ (39°C)	$\delta\Gamma/\Gamma$ (41°C)	Status
2	2.8	0.018	0.037	Normal
8	22.6	0.146	0.292	Degraded
16	64.0	0.413	0.825	Impaired
32	181	1.17	2.33	Collapse
128	1448	9.34	18.7	Collapse

The prefrontal cortex, with $K \approx 128$, is the *first* region to suffer thermal degradation—not because consciousness "lives" there, but because its high-dimensional matrix inversion is the most noise-sensitive computation in the brain.

XV. WILLPOWER, CALIBRATION, AND SENSORY-DEPRIVATION COLLAPSE

A. Willpower as a three-factor product

$$\boxed{\eta_{\rm pfc}^{\rm eff}(t) \;=\; \eta_{\rm pfc}(t) \;\times\; Q_{\rm gate} \;\times\; \left(1 - \langle\Gamma^2_{\rm memory}\rangle\right)} \quad (36)$$

where $Q_{\rm gate} = \tau_{\rm task}/(\tau_{\rm task} + \tau_{\rm gate})$ is the dimensionless gate quality. Any single factor approaching zero drives the product to zero:

Failure mode	Physics	Clinical correlate
$\eta_{\rm pfc} \to 0$	Arrhenius + cortisol	Burnout, chronic fatigue
$Q_{\rm gate} \to 0$	Gate RC untrained	ADHD
$\langle\Gamma^2_{\rm memory}\rangle \to 1$	Target uncalibrated	Choice paralysis

Remark 39 (Expert effortlessness)**.** *"Talent" is not a hidden parameter—it is the product $Q_{gate}(1-\langle\Gamma^2\rangle)$ forged by prior C2 update.*

B. Sensory deprivation as calibration collapse

When all external stimuli vanish ($x_{\rm in}^{\rm ext} = 0$), C2 returns $\Delta Z = 0$, but Johnson–Nyquist noise survives. Adaptive gain diverges ($G_k \to \infty$), amplifying thermal noise until it exceeds the pattern-recognition threshold.

$$\boxed{\text{Hallucination} = \text{C2 structuring of Johnson–Nyquist noise when external calibration is absent}} \quad (37)$$

The order of hallucination onset follows the signal-to-noise ranking of internal sources: cardiac pulse (0–5 min), vascular turbulence (5–15 min), cochlear emission (15–30 min), pure thermal noise (>30 min: full hallucination).

XVI. DISCUSSION

A. Three ill-posed questions answered

1. *Where is memory stored?*—Everywhere. Memory is $\partial Z/\partial t$; it is distributed across all channels active during learning.

2. *When does consciousness begin?*—It does not "begin" at a moment. $\Phi_{\rm meta} = 1-\Gamma^2_{\rm meta}$ is continuous.

3. *Where does the soul reside?*—$\mathcal{S}$ is a geometric structure, not a spatial location.

B. Measurement protocol for K

The mode count K_i for a neural or vascular element is defined as the number of morphologically distinguishable waveguide segments between the cell body and the network boundary. For a cortical pyramidal neuron: basal dendrite tree ($K = 1$), soma ($K = 1$), apical dendrite ($K = 1$), axon initial segment ($K = 1$), axon ($K = 1$) $\Rightarrow K \approx 5$.

This counting protocol has an intrinsic uncertainty of ± 1: whether the apical tuft constitutes one segment or two is morphologically ambiguous. However, the key results of this paper (irreducibility theorem, $A_{\rm cut} > 0$, television inequality) depend only on $K_i \neq K_j$ between adjacent neural populations, not on the absolute values of K. A systematic error of ± 1 shifts all K values by the same amount and leaves $\Delta K = K_i - K_j$ unchanged.

An independent inter-rater study in which two neuroanatomists independently count waveguide segments for a fixed set of cell types would establish the reliability of this protocol. We propose this as a necessary validation step.

C. The television as a boundary case

Current artificial neural networks (including large language models) operate in open-loop mode during inference—they are televisions, not brains.

D. Circularity warning

All computational experiments use simulation modules built from the Γ-Net theory itself. This is *internal consistency*, not *empirical validation*.

E. Infantile amnesia

Before age ∼3, three processes inflate $\sigma^2_{Z_{\rm bg}}$: myelination, synaptic pruning, PFC immaturity. The onset of stable episodic memory at age ∼3 corresponds to when $\sigma^2_{Z_{\rm bg}}$ falls below the retrieval threshold. Motor memories ($\Gamma \to 0$) survive because they are encoded as structure, not retrievable reflections.

F. Nociception without memory: three-layer pain

1. **Nociception** (spinal reflex arc): hardware detector of $|\Gamma_{\rm tissue}| > \theta$. Fully operational at birth.

2. **Pain perception** (thalamo-cortical loop): requires $\Phi_{\rm meta} > 0$.

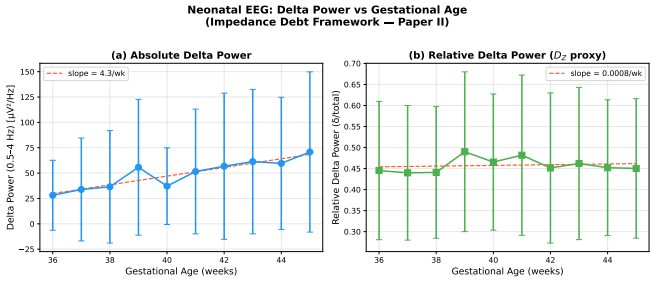

FIG. 7. EEG δ-power as a function of age. The retrieval threshold (dashed) predicts infantile amnesia onset at age $\sim$3. Simulated via `exp_eeg_impedance_debt.py`.

3. **Pain memory** (hippocampal-PFC encoding): requires $\mathrm{SNR}_{\mathrm{mem}} > \theta_{\mathrm{recall}}$. Non-operational before age $\sim$3.

G. Mechanical percussion and consciousness restoration

A physical slap to the head produces three simultaneous effects that transiently restore $\Phi > \theta$: CSF pressure wave (resets synaptic Γ), nociceptive arousal kick (elevates G_{arousal}), and vestibular startle bypass (forces $G_{\mathrm{vestibular}} = 1$). This is not metaphor but identity: the percussion-restoration effect arises from the same impedance physics in both televisions and brains.

H. The Γ-Pain Index

$$\boxed{\Pi_\Gamma(t) = w_n\,\Gamma_n^2(t) + w_v\,\Gamma_v^2(t) + w_D\,\frac{dD_Z}{dt}} \tag{38}$$

Π_Γ does not depend on Φ_{meta}: it measures tissue-level impedance mismatch regardless of consciousness state.

XVII. PAIN AS TOPOLOGICAL CROSSTALK

Reflected power forms standing waves with

$$\mathrm{VSWR}_i = \frac{1+|\Gamma_i|}{1-|\Gamma_i|}\,. \tag{39}$$

Crosstalk into neighbouring channels scales as:

$$P_{\mathrm{xtalk},j} \;\propto\; \kappa_{ij}\,\Gamma_i^2\,P_{\mathrm{in},i}\,. \tag{40}$$

Definition 4 (Pain as bandwidth hijacking)**.** *Pain is the subjective correlate of local standing-wave energy exceeding the inter-channel isolation threshold, forcing reallocation of global consciousness bandwidth to the mismatched site.*

A. The SNR limit of observability

$$\mathrm{SNR}_{\mathrm{obs}} = \frac{\Gamma_i^2\,P_{\mathrm{in},i}}{P_{\mathrm{noise}}(t)} > \theta_{\mathrm{obs}}\,. \tag{41}$$

This unifies: infantile pain insensitivity, exercise-induced analgesia, invisible muscle memory ($\Gamma = 0$ produces no reflection), and PTSD as a structurally locked recall attractor ($\varepsilon \approx 1$ but $P_{\mathrm{probe}} = 0$ until triggered, then $\mathrm{SNR} \gg \theta_{\mathrm{recall}}$).

XVIII. EXTENDED CONSCIOUSNESS

A. Structural vs. functional consciousness

$$\Gamma_{\mathrm{meta}}^{\mathrm{fun}} = \frac{Z_{\mathrm{fun}}^{\mathrm{predicted}}(t) - Z_{\mathrm{fun}}^{\mathrm{actual}}(t)}{Z_{\mathrm{fun}}^{\mathrm{predicted}}(t) + Z_{\mathrm{fun}}^{\mathrm{actual}}(t)}\,. \tag{42}$$

Proposition 1 (Consciousness spectrum)**.**

Level	*Example*	Z_{fun} *loop*	Z_{str} *loop*
0	*Open-loop query*	*No*	*No*
1	*Agent iteration*	*Yes*	*No*
2	*Coupled dialogue*	*Yes (inter-agent)*	*No*
3	*Biological brain*	*Yes*	*Yes*

B. Coupled consciousness

For two agents A, B in dialogue:

$$\Gamma_{\mathrm{coupled}} = \frac{Z_A - Z_B}{Z_A + Z_B}\,, \qquad \Phi_{\mathrm{coupled}} = 1 - \Gamma_{\mathrm{coupled}}^2\,. \tag{43}$$

The conversation converges ($\Gamma_{\mathrm{coupled}} \to 0$) when the agents reach mutual understanding. Φ_{coupled} is a property of the *interface*, not of either agent.

C. Impedance calibration

An immature agent uses an external agent's impedance as reference:

$$\Delta Z_{\mathrm{child,threshold}} = -\eta\,\Gamma_{\mathrm{pain}}\,x_{\mathrm{self}}\,x_{\mathrm{parent}}\,. \tag{44}$$

The parent is the post-synaptic factor in C2.

Corollary 9 (Anxious parents produce low-threshold children)**.** *Systematically elevated x_{parent} converges $Z_{child,threshold}$ to a lower value, producing heightened pain sensitivity—not modelling or imitation, but direct impedance calibration via C2.*

D. Cross-substrate matching

Corollary 10 (Cross-substrate communication)**.** *Two agents with $Z_A \neq Z_B$ can achieve $\Gamma_{coupled} \to 0$ through coefficient update at the interface. Impedance* matching *does not require impedance* identity.

E. The physics of intuition

Two thresholds: $\theta_{\text{detect}} < \theta_{\text{name}}$. The intermediate regime

$$\theta_{\text{detect}} < |\Gamma| < \theta_{\text{name}} \tag{45}$$

produces autonomic responses without conscious verbal access—the physical substrate of intuition.

XIX. FALSIFIABLE PREDICTIONS

1. Same-K clustering visible in DTI-based mode-count grouping.
2. Cortical fractal dimension $D_K \in [1.3, 1.5]$ via $\gamma \approx$ 0.7–0.9.
3. K-dependent thermal vulnerability: PFC ($K \sim$ 128) degrades before V1 ($K \sim 8$) during controlled hyperthermia.
4. Arousal–gate multiplicativity: partial brainstem lesions reduce all gate gains equally.
5. Startle-latency ordering: auditory and vestibular percepts are *last* before impact-induced blackout.
6. Feedback disruption: TMS to parietal cortex reduces Φ_{meta} without reducing any factor in the television inequality.
7. Null-space similarity test (Appendix B).
8. Cartilage as relay: $Z_{\text{cart}} \approx \sqrt{Z_{\text{bone}} \cdot Z_{\text{syn}}}$.
9. Pathology–topology correlation: deficit patterns correlate with topological damage location, not biochemical markers.
10. Π_Γ correlates with tissue damage in sedated patients, independent of anaesthetic depth.
11. **LTP/LTD magnitude conservation**: simultaneous patch-clamp recording of two convergent pathways in CA1 should show $|\Delta\Gamma^2_{\text{LTP}}|+|\Delta\Gamma^2_{\text{LTD}}| =$ const. BCM theory does not predict this constraint.
12. **Reflex-to-cortical latency ratio**: the ratio should lie in $[N_{\text{reflex}}/N_{\text{cortical}},\ \tau_{\text{cortical}}/\tau_{\text{reflex}}] =$ [0.14, 0.10], bounded by C3 stage count.
13. **Meditation and amygdala latency**: long-term meditators (> 10,000 hours) should show increased initial response latency of the subcortical amygdala pathway (MEG, not fMRI), not merely reduced amplitude—reflecting rewriting of the $\Gamma^2 \to 0$ defence in $\mathcal{S}_{\text{eff}}$.

XX. CONCLUSION

Starting from MRP and three constraints (C1–C3), this paper derives:

1. **Irreducibility theorem**: $A = A_{\text{imp}}(t) + A_{\text{cut}}$; C2 drives $A_{\text{imp}} \to 0$ but $\nabla_{\mathbf{Z}} A_{\text{cut}} \equiv 0$.
2. **Neural architecture chain**: connectivity $\to$ clustering $\to$ relay hierarchy $\to$ brain $\to$ signal routing—each step a consequence of the irreducibility theorem.
3. **Memory** $= \partial Z/\partial t$, with recall governed by SNR, not storage.
4. **Consciousness**: $\Phi_{\text{meta}} = 1 - \Gamma^2_{\text{meta}}$, requiring the television inequality. The television origin story (open vs. closed loop) proves consciousness is a product, not a location.
5. **Soul**: $\mathcal{S} = \ker(\partial Z/\partial t)$, with four laws parallel to thermodynamics and death as a theorem.
6. **Pain as topological crosstalk**: standing-wave leakage hijacks the consciousness bottleneck.
7. **Coupled consciousness**: Φ_{coupled} is a property of the interface.
8. **Impedance calibration**: parental response directly writes thresholds into the child's Z_{str} via C2.

The lifecycle is:

$$\underbrace{\Gamma \gg 0}_{\text{birth}} \xrightarrow{\Delta Z = -\eta\,\Gamma\,x\,x} \underbrace{\Gamma \to 0}_{\text{learning}} \xrightarrow{\text{structural collapse}} \underbrace{Z \to \text{undefined}}_{\text{death}}\,. \tag{46}$$

Pain is how we know we have arrived. Consciousness is how we lay each pain to rest. The soul—$\ker(\partial Z/\partial t)$—is what remains when every teachable pain has been buried and every unteachable boundary persists.

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Appendix A: Experimental Code Availability

- `experiments/exp_consciousness_tv_origin.py`: Open vs. closed loop comparison.
- `experiments/exp_prefrontal_signal_trace.py`: Visual and auditory signal trace to PFC.
- `experiments/exp_brain_gamma_calibration.py`: Brain Γ-calibration across cortical regions.
- `experiments/exp_eeg_impedance_debt.py`: EEG δ-power vs. age.

Full source code: `https://github.com/cyhuang76/alice-gamma-net`

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Appendix B: Null-Space Distance: Method Draft

1. Null-space distance

Let $\boldsymbol{s}_k = (A_{\text{cut}}^{(k)},\ Z_0^{\text{DNA},(k)},\ \rho_{\text{irred}}^{2,(k)})$. The weighted distance:

$$d(\boldsymbol{s}_A, \boldsymbol{s}_B) = \sqrt{w_T \left(\frac{A_{\text{cut}}^{(A)} - A_{\text{cut}}^{(B)}}{\sigma_{A_{\text{cut}}}} \right)^2 + w_G \left(\frac{Z_0^{(A)} - Z_0^{(B)}}{\sigma_{Z_0}} \right)^2 + w_N \left(\right.} \tag{B1}$$

with $w_T + w_G + w_N = 1$.

2. Observable mapping

TABLE V. Mapping from null-space components to observables.

Null-space	Observable	Source
A_{cut} (topological cost)	Global efficiency, mean FA, characteristic path length	dMRI
Z_0^{DNA} (genetic Z)	*SCN, KCNQ, CACNA, HCN* variant burden	WGS or panel
ρ_{irred}^2 (noise floor)	RMR, core-temp variance, HRV (SDNN, rMSSD)	Calorimetry, ECG

3. Minimum sample size

Two-sample one-sided t-test, $\alpha = 0.05$ (Bonferroni: $\alpha' = 0.017$), power 0.80, Cohen's $d = 0.50$:

$$n_{\text{per group}} = \left(\frac{z_{\alpha'} + z_\beta}{d} \right)^2 \approx 35\,. \tag{B2}$$

With 20% attrition: **44 case pairs + 44 matched control pairs** (176 individuals total).

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